Supplementary file

Differences in gastrointestinal hormones and appetite ratings between individuals with and without obesity – A systematic review and meta-analysis

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Contents

Search strategy: 3

MEDLINE 3

Embase 5

PsycINFO 7

Cochrane Library 8

Web of Science 10

ClinicalTrials.gov 10

Flow diagram 12

Summary of contributing articles 13

Active ghrelin (basal) 13

Total ghrelin (basal) 15

Active GLP-1 (basal) 17

Total GLP-1 (basal) 19

Active PYY (basal) 21

Total PYY (basal) 23

CKK (basal) 25

Total ghrelin (AUC) 27

Active GLP-1 (AUC) 29

Total PYY (AUC) 31

Hunger (fasting) 33

Fullness (fasting) 35

Desire to eat (DTE) (fasting) 37

Prospective food consumption (PFC) (fasting) 39

Hunger (AUC) 41

Fullness (AUC) 43

Desire to eat (DTE) (AUC) 45

Prospective food consumption (PFC) (AUC) 47

Supplementary table 19 – Risk of bias assessment 49

Supplementary table 20. Study characteristics 54

Search strategy:

MEDLINE

Ovid MEDLINE(R) ALL <1946 to September 01, 2021>

#	Searches
1	exp Obesity/
2	exp Obesity, Morbid/
3	exp Obesity, Abdominal/
4	exp Overweight/
5	adipos*.ti,ab,kw.
6	obes*.ti,ab,kw.
7	overweight*.ti,ab,kw.
8	over-weight*.ti,ab,kw.
9	((high* or unhealthy) adj3 (body-mass-ind* or BMI?)).ti,ab,kw.
10	or/1-9 [Combined thesaurus and text words for overweight/obesity concept]
11	exp Ideal Body Weight/
12	normal-weight*.ti,ab,kw.
13	normal-body-weight*.ti,ab,kw.
14	normal-body-mass*.ti,ab,kw.
15	healthy-weight*.ti,ab,kw.
16	healthy-body-weight*.ti,ab,kw.
17	healthy-body-mass*.ti,ab,kw.
18	non-obese.ti,ab,kw.
19	nonobese.ti,ab,kw.
20	non-overweight*.ti,ab,kw.
21	slim.ti,ab,kw.
22	lean.ti,ab,kw.
23	((normal or control? or healthy or low*) adj3 (overweight or over-weight or obese* or men or women or subject* or person* or body-mass-ind* or BMI? or volunteer*)).ti,ab,kw.
24	or/11-23 [Combined thesaurus and text words for healthy weight concept]
25	exp Hunger/
26	exp Appetite/
27	exp Appetite Regulation/
28	exp Satiation/
29	appetite?.ti,ab,kw.
30	apetite?.ti,ab,kw.
31	hunger.ti,ab,kw.

32 fullness.ti,ab,kw.
33 satiation.ti,ab,kw.
34 satiety.ti,ab,kw.
35 prospective-food-consumption*.ti,ab,kw.
36 ((desire or craving*) adj3 (eat* or food or feed*)).ti,ab,kw.
37 exp Ghrelin/
38 ghrelin.ti,ab,kw.
39 GHRL.ti,ab,kw.
40 appetite-regulating-hormone.ti,ab,kw.
41 exp Cholecystokinin/
42 cholecystokinin.ti,ab,kw.
43 CCK.ti,ab,kw.
44 uropancreozymin.ti,ab,kw.
45 pancreozymin.ti,ab,kw.
46 exp Peptide-YY/
47 peptide-YY.ti,ab,kw.
48 PYY.ti,ab,kw.
49 PYY3-36.ti,ab,kw.
50 peptide-tyrosine-tyrosine.ti,ab,kw.
51 exp Glucagon-Like Peptide 1/
52 glucagon-like-peptide-1.ti,ab,kw.
53 GLP-1.ti,ab,kw.
54 GLP1.ti,ab,kw.
55 or/25-54 [Combined thesaurus and text words for appetite/appetite markers concept]
56 and/10,24,55
57 56 not (comment or editorial or news or newspaper article).pt. [Excluded publication types]
58 (Animals/ or Models, Animal/ or Disease Models, Animal/) not Humans/
59 ((animal or animals or canine* or dog or dogs or feline or hamster* or lamb or lambs or mice or monkey or monkeys or mouse or murine or pig or pigs or piglet* or porcine or primate* or rabbit* or rat or rats or rodent* or sheep* or veterinar*) not (human* or patient*)).ti,kf,jw.
60 57 not (58 or 59) [Animal-only studies excluded]
61 (exp infant/ or exp child/ or exp adolescent/) not exp adult/
62 ((child* or stepchild* or step-child* or kid or kids or girl or girls or boy or boys or teen* or youth* or youngster* or adolescent* or adolescence or preschool* or pre-school* or kindergarten* or school* or juvenile* or minors or p?ediatric* or PICU) not adult*).ti,ab.
63 60 not (61 or 62) [Non-adults only studies excluded]
64 limit 63 to (english or norwegian or swedish or danish or portuguese or french or spanish) [Limiting results to English, Norwegian, Swedish, Danish, Portugese, French or Spanish language]

Embase

Embase <1974 to 2021 September 01>

#	Searches
1	exp Obesity/
2	exp Morbid Obesity/
3	exp Abdominal Obesity/
4	exp Obesity Management/
5	exp Maternal Obesity/
6	adipos*.ti,ab,kw.
7	obes*.ti,ab,kw.
8	overweight*.ti,ab,kw.
9	over-weight*.ti,ab,kw.
10	((high* or unhealthy) adj3 (body-mass-ind* or BMI?)).ti,ab,kw.
11	or/1-10 [Combined thesaurus and text words for overweight/obesity concept]
12	exp Ideal Body Weight/
13	normal-weight*.ti,ab,kw.
14	normal-body-weight*.ti,ab,kw.
15	normal-body-mass*.ti,ab,kw.
16	healthy-weight*.ti,ab,kw.
17	healthy-body-weight*.ti,ab,kw.
18	healthy-body-mass*.ti,ab,kw.
19	non-obese.ti,ab,kw.
20	nonobese.ti,ab,kw.
21	non-overweight*.ti,ab,kw.
22	slim.ti,ab,kw.
23	lean.ti,ab,kw.
24	((normal or control? or healthy or low*) adj3 (overweight or over-weight or obese* or men or women or subject* or person* or body-mass-ind* or BMI? or volunteer*)).ti,ab,kw.
25	or/12-24
26	or/12-24 [Combined thesaurus and text words for healthy weight concept]
27	exp Hunger/
28	exp Appetite/
29	exp Decreased Appetite/
30	exp "Loss Of Appetite"/
31	exp Appetite Disorder/
32	exp Appetite Stimulant/
33	exp Increased Appetite/
34	exp Satiety/
35	appetite?.ti,ab,kw.

36 apetite?.ti,ab,kw.
37 hunger.ti,ab,kw.
38 fullness.ti,ab,kw.
39 satiation.ti,ab,kw.
40 satiety.ti,ab,kw.
41 prospective-food-consumption*.ti,ab,kw.
42 ((desire or craving*) adj3 (eat* or food or feed*)).ti,ab,kw.
43 exp Ghrelin/
44 ghrelin.ti,ab,kw.
45 GHRL.ti,ab,kw.
46 appetite-regulating-hormone.ti,ab,kw.
47 exp Cholecystokinin/
48 cholecystokinin.ti,ab,kw.
49 CCK.ti,ab,kw.
50 uropancreozymin.ti,ab,kw.
51 pancreozymin.ti,ab,kw.
52 exp Peptide-YY/
53 peptide-YY.ti,ab,kw.
54 PYY.ti,ab,kw.
55 PYY3-36.ti,ab,kw.
56 peptide-tyrosine-tyrosine.ti,ab,kw.
57 exp Glucagon-Like Peptide 1/
58 glucagon-like-peptide-1.ti,ab,kw.
59 GLP-1.ti,ab,kw.
60 GLP1.ti,ab,kw.
61 or/27-60 [Combined Combined thesaurus and text words for appetite/appetite markers concept]
62 and/11,26,61
63 62 not (comment or editorial or news or newspaper article).pt. [Exluded publication types]
64 (exp Animal/ or exp Juvenile Animal/ or Adult Animal/ or Animal Cell/ or Animal Tissue/ or Nonhuman/ or Animal Experiment/ or Animal Model/) not Human/
65 (animal or animals or canine* or dog or dogs or feline or hamster* or lamb or lambs or mice or monkey or monkeys or mouse or murine or pig or pigs or piglet* or porcine or primate* or rabbit* or rat or rats or rodent* or sheep* or veterinar*).ti,kw,dq,jx. not (human* or patient*).mp.
66 63 not (64 or 65) [Animal-only studies excluded]
67 (exp Infant/ or exp Child/ or exp Adolescent/) not exp Adult/
68 ((child* or stepchild* or step-child* or kid or kids or girl or girls or boy or boys or teen* or youth* or youngster* or adolescent* or adolescence or preschool* or pre-school* or kindergarten* or school* or juvenile* or minors or p?ediatric* or PICU) not adult*).ti,ab.
69 66 not (67 or 68) [non-adults only studies excluded]

70	limit 69 to (english or norwegian or swedish or danish or portuguese or french or spanish) [Limiting results to English, Norwegian, Swedish, Danish, Portugese, French or Spanish language]
71	limit 70 to medline
72	70 not (medline or "pubmed not medline").ns. [Excluding MEDLINE and PubMed not MEDLINE records]

PsycINFO

APA PsycInfo <1806 to August Week 4 2021>

#	Searches
1	exp Obesity/
2	exp Overweight/
3	adipos*.tw.
4	obes*.tw.
5	overweight*.tw.
6	over-weight*.tw.
7	((high* or unhealthy) adj3 (body-mass-ind* or BMI?)).tw.
8	or/1-7 [Combined thesaurus and text words for overweight/obesity concept]
9	normal-weight*.tw.
10	normal-body-weight*.tw.
11	normal-body-mass*.tw.
12	healthy-weight*.tw.
13	healthy-body-weight*.tw.
14	healthy-body-mass*.tw.
15	non-obese.tw.
16	nonobese.tw.
17	non-overweight*.tw.
18	slim.tw.
19	lean.tw.
20	((normal or control? or healthy or low*) adj3 (overweight or over-weight or obese* or men or women or subject* or person* or body-mass-ind* or BMI? or volunteer*)).tw.
21	or/9-20 [Combined thesaurus and text words for healthy weight concept]
22	exp Hunger/
23	exp Appetite/
24	exp Satiation/
25	appetite?.tw.
26	apetite?.tw.
27	hunger.tw.
28	fullness.tw.

29	satiation.tw.
30	satiety.tw.
31	prospective-food-consumption*.tw.
32	((desire or craving*) adj3 (eat* or food or feed*)).tw.
33	exp Ghrelin/
34	ghrelin.tw.
35	GHRL.tw.
36	appetite-regulating-hormone.tw.
37	exp Cholecystokinin/
38	cholecystokinin.tw.
39	CCK.tw.
40	uropancreozymin.tw.
41	pancreozymin.tw.
42	peptide-YY.tw.
43	PYY.tw.
44	PYY3-36.tw.
45	peptide-tyrosine-tyrosine.tw.
46	glucagon-like-peptide-1.tw.
47	GLP-1.tw.
48	GLP1.tw.
49	or/22-48 [Combined thesaurus and text words for appetite/appetite markers]
50	and/8,21,49
51	50 not ("column opinion" or "comment reply" or "Editorial").dt. [Excluded document types]
52	((human or animal) not human).po.
53	(animal or animals or canine* or dog or dogs or feline or hamster* or lamb or lambs or mice or monkey or monkeys or mouse or murine or pig or pigs or piglet* or porcine or primate* or rabbit* or rat or rats or rodent* or sheep* or veterinar*).ti,ab. not (human* or patient*).mp.
54	51 not (52 or 53) [Animal-only studies excluded]
55	((adolescence 13 17 yrs or adulthood 18 yrs older or childhood birth 12 yrs) not adulthood 18 yrs older).ag.
56	((child* or stepchild* or step-child* or kid or kids or girl or girls or boy or boys or teen* or youth* or youngster* or adolescent* or adolescence or preschool* or pre-school* or kindergarten* or school* or juvenile* or minors or p?ediatric* or PICU) not adult*).ti,ab.
57	54 not (55 or 56) [non-adult studies excluded]
58	limit 57 to (english or norwegian or swedish or danish or portuguese or french or spanish) [Limiting results to English, Norwegian, Swedish, Danish, Portugese, French or Spanish language]

Cochrane Library

ID	Search	Hits

#1 (adipos*):ti,ab,kw
#2 (obes*):ti,ab,kw
#3 (overweight*):ti,ab,kw
#4 (over-weight*):ti,ab,kw
#5 ((high* or unhealthy) NEAR/3 (body-mass-ind* or BMI?)):ti,ab,kw
#6 #1 OR #2 OR #3 OR #4 OR #5
#7 (normal-weight*):ti,ab,kw
#8 (normal-body-weight*):ti,ab,kw
#9 (normal-body-mass*):ti,ab,kw
#10 (healthy-weight*):ti,ab,kw
#11 (healthy-body-weight*):ti,ab,kw
#12 (healthy-body-mass*):ti,ab,kw
#13 (non-obese):ti,ab,kw
#14 (nonobese):ti,ab,kw
#15 (non-overweight*):ti,ab,kw
#16 (slim):ti,ab,kw
#17 (lean):ti,ab,kw
#18 ((normal or control? or healthy or low*) NEAR/3 (overweight or over-weight or obese* or men or women or subject* or person* or body-mass-ind* or BMI? or volunteer*)):ti,ab,kw
#19 #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18
#20 (appetite?):ti,ab,kw
#21 (apetite?):ti,ab,kw
#22 (hunger):ti,ab,kw
#23 (fullness):ti,ab,kw
#24 (satiation):ti,ab,kw
#25 (satiety):ti,ab,kw
#26 (prospective-food-consumption*):ti,ab,kw
#27 ((desire or craving*) NEAR/3 (eat* or food or feed*)):ti,ab,kw
#28 (ghrelin):ti,ab,kw
#29 (GHRL):ti,ab,kw
#30 (appetite-regulating-hormone):ti,ab,kw
#31 (cholecystokinin):ti,ab,kw
#32 (CCK):ti,ab,kw
#33 (uropancreozymin):ti,ab,kw
#34 (pancreozymin):ti,ab,kw
#35 (peptide-YY):ti,ab,kw
#36 (PYY):ti,ab,kw
#37 ("PYY3-36"):ti,ab,kw
#38 (peptide-tyrosine-tyrosine):ti,ab,kw
#39 (glucagon-like-peptide-1):ti,ab,kw
#40 (GLP-1):ti,ab,kw
#41 (GLP1):ti,ab,kw
#42 #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41
#43 #6 AND #19 AND #42
#44 ((child* OR stepchild* OR step-child* OR kid OR kids OR girl OR girls OR boy OR boys OR teen* OR youth* OR youngster* OR adolescent* OR adolescence OR preschool* OR pre-school* OR kindergarten* OR school* OR juvenile* OR minors OR p?ediatric* OR PICU) NOT adult*):ti,ab,kw

#45 #43 NOT #44

Web of Science

Web of Science Core Collection

16 #13 not #14 and Editorial Materials (Exclude – Document Types)
15 #13 not #14
14 TS=(("child*" or "stepchild*" or "step-child*" or "kid" or "kids" or "girl" or "girls" or "boy" or "boys" or "teen*" or "youth*" or "youngster*" or "adolescent*" or "adolescence" or "preschool*" or "pre-school*" or "kindergarten*" or "school*" or "juvenile*" or "minors" or "p?ediatric*" or "PICU") not "adult*")
13 #11 not #12
12 TS=(("animal" or "animals" or "canine*" or "dog" or "dogs" or "feline" or "hamster*" or "lamb" or "lambs" or "mice" or "monkey" or "monkeys" or "mouse" or "murine" or "pig" or "pigs" or "piglet*" or "porcine" or "primate*" or "rabbit*" or "rat" or "rats" or "rodent*" or "sheep*" or "veterinar*") not ("human*" or "patient*"))
11 #10 AND #6 AND #3
10 #9 OR #8 OR #7
9 TS=("ghrelin" or "GHRL" or "appetite-regulating-hormone" or "cholecystokinin" or "CCK" or "uropancreozymin" or "pancreozymin" or "peptide-YY" or "PYY" or "PYY3-36" or "peptide-tyrosine-tyrosine" or "glucagon-like-peptide-1" or "GLP-1" or "GLP1")
8 TS=(("desire" or "craving*") NEAR/3 ("eat*" or "food" or "feed*"))
7 TS=("appetite?" or "apetite?" or "hunger" or "fullness" or "satiation" or "satiety" or "prospective-food-consumption*")
6 #4 OR #5
5 TS=((normal or control? or healthy or low*) NEAR/3 (overweight or over-weight or obese* or men or women or subject* or person* or body-mass-ind* or BMI? or volunteer*))
4 TS=("normal-weight*" or "normal-body-weight*" or "normal-body-mass*" or "healthy-weight*" or "healthy-body-weight*" or "healthy-body-mass*" or "non-obese" or "nonobese" or "non-overweight*" or "slim" or "lean")
3 #2 OR #1
2 TS=(("high*" or "unhealthy") NEAR/3 ("body-mass-ind*" or "BMI?"))
1 TS=("adipos*" or "obes*" or "overweight*" or "over-weight*")

ClinicalTrials.gov

(hunger OR appetite OR fullness OR satiation OR satiety OR prospective-food-consumption OR desire OR craving OR cravings OR ghrelin OR GHRL OR appetite-regulating-hormone OR cholecystokinin OR PYY OR PYY3-36 OR GLP-1 OR GLP1) AND (obese OR obesity OR overweight OR over-weight OR body-mass-index OR BMI OR overweight) AND (normal-weight OR normal-body-weight OR normal-body-mass OR healthy-weight OR healthy-body-weight OR healthy-body-mass OR non-obese OR nonobese OR lean OR slim) AND (Adult OR Older Adult)

Flow diagram

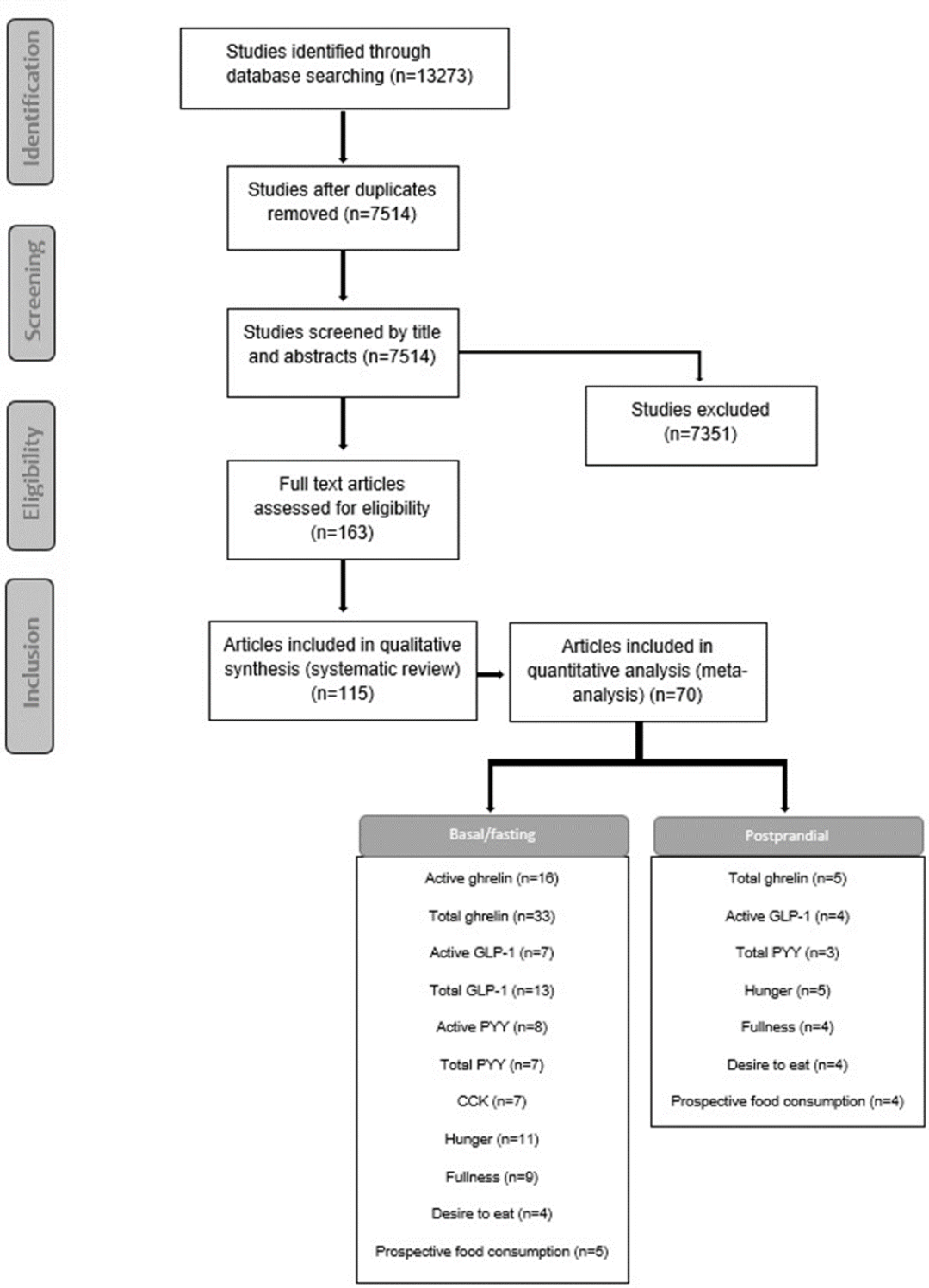

Figure 1: Flow diagram of the selection process.

Summary of contributing articles

Basal hormone concentrations

Active ghrelin (basal)

Table 1: Summary of contributing articles for active ghrelin (basal)

		Obesity group				Control group			
Author (year)	**Study design**	**n**	**% female**	**Age (years)**	**BMI (kg/m²)**	**n**	**% female**	**Age (years)**	**BMI (kg/m²)**
Andarini, 2017 [47]	Cross-sectional	16	0.0	21.4 (1.9)	33.6 (4.8)	16	0.0	20.6 (1.1)	21.3 (1.1)
Arafat, 2013 [26]	Parallel RCT	11	55.0	28.4 (8.6)	34.4 (5.6)	13	54.0	25.1 (2.2)	21.7 (2.2)
Carrol, 2007 [40]	Cross-sectional	20	55.0	47.2 (11.0)	36.8 (4.8)	19	47.0	39.2 (10.3)	22.6 (1.2)
Chearskul, 2012 [41]	Cross-sectional	33	100.0	33.6 (6.3)	29.4 (3.6)	53	100.0	30.1 (5.8)	20.7 (2.2)
Dardzinska, 2014 [27]	.	12	71.0	35.4 (9.1)	43.8 (6.8)	12	92.0	37.2 (9.4)	23.0 (3.5)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Douglas, 2017 [42]	Crossover	25	50.0	45.0 (12.4)	29.2 (2.9)	22	44.0	37.5 (15.2)	22.4 (1.5)
Heden, 2013 [97]	Crossover	14	57.0	25.1 (5.0)	34.8 (4.4)	14	43.0	26.0 (6.0)	22.9 (1.7)
Homaee, 2011 [30]	Cross-sectional	19	0.0	27.5 (5.8)	31.0 (3.6)	19	0.0	26.9 (5.6)	18.5 (2.2)
Iceta, 2019 [31]	Cross-sectional	55	100.0	38.0 (11.1)	41.5 (5.9)	29	100.0	37.0 (10.8)	21.5 (2.2)
Karcz-Socha, 2011 [45]	Cross-sectional	96	50.0	51.5 (6.5)	35.3 (2.9)	46	52.0	51.2 (6.5)	23.4 (1.5)
Kolodziejski, 2018 [43]	Cross-sectional	15	100.0	53.0 (8.5)	39.8 (1.0)	15	100.0	58.6 (10.3)	22.3 (0.5)
Korek, 2013 [32]	Cross-sectional	19	90.0	42.2 (3.3)	34.7 (4.9)	17	89.0	42.9 (5.3)	21.1 (1.9)
Krzyzanowska-Swinirska, 2007 [46]	Cross-sectional	30	100.0	44.3 (4.3)	34.4 (4.1)	32	100.0	30.6 (3.6)	21.3 (1.7)
Lopez-Aguilar, 2018 [33]	Cross-sectional	50	52.0	29.2 (6.3)	34.5 (5.2)	80	67.0	26.4 (5.6)	22.7 (1.5)
Marzullo, 2004 [35]	Cross-sectional	20	50.0	31.8 (2.5)	41.3 (1.1)	20	50.0	33.5 (2.4)	22.4 (0.6)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

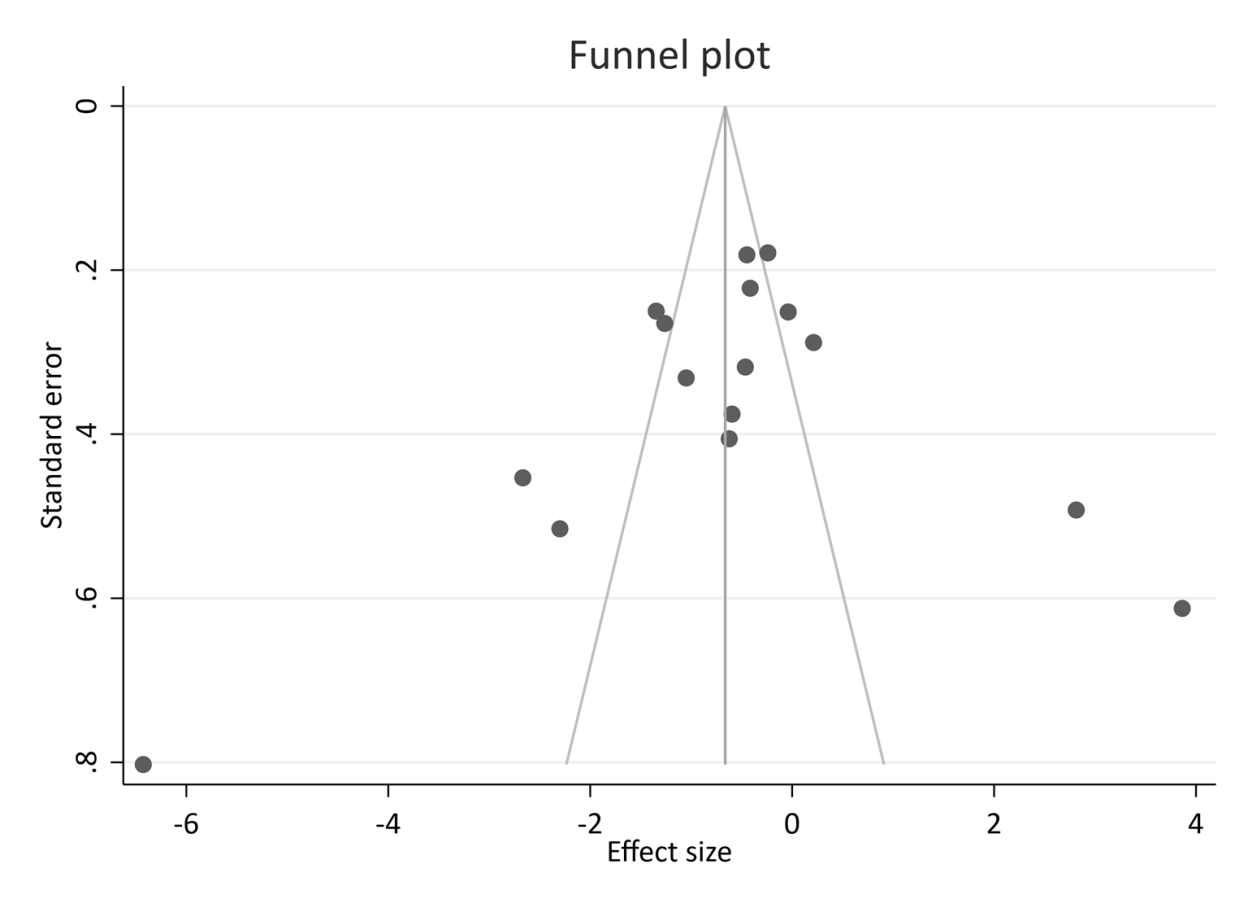

Figure 2: Funnel plot for active ghrelin (basal).

Total ghrelin (basal)

Table 2: Summary of contributing articles for total ghrelin (basal)

Author (year)	Study design	Obesity group n	Obesity group % female	Obesity group Age (years)	Obesity group BMI (kg/m^2)	Control group n	Control group % female	Control group Age (years)	Control group BMI (kg/m^2)
Acosta, 2015 [14]	Cross-sectional	201	72.0	37.8 (12.1)	35.4 (3.9)	105	61.0	37.3 (12.0)	27.6 (0.1)
Arafat, 2013 [26]	Parallel RCT	11	55.0	28.4 (8.6)	34.4 (5.6)	13	54.0	25.1 (2.2)	21.7 (2.2)
Batterham, 2003 [49]	Crossover	12	50.0	29.0 (8.3)	33.0 (3.1)	12	50.0	27.3 (1.4)	20.5 (0.3)
Bogdanov, 2019 [51]	Cross-sectional	15	80.0	38.7 (9.4)	41.2 (2.1)	15	87.0	37.0 (7.1)	21.8 (2.6)
Bowen, 2006 [52]	Crossover	47	0.0	56.8 (7.5)	30.1 (3.4)	25	0.0	50.5 (12.0)	23.3 (1.0)
Brownley, 2012 [53]	Crossover	20	50.0	34.9 (9.0)	34.7 (2.8)	20	50.0	27.9 (7.0)	22.9 (1.4)
Carlson, 2009 [54]	Cross-sectional	13	100.0	35.6 (9.7)	44.5 (7.1)	10	100.0	32.2 (8.6)	23.1 (1.3)
Cassar, 2015 [55]	Cross-sectional	18	100.0	35.0 (5.0)	31.0 (3.0)	22	100.0	28.0 (6.0)	22.0 (2.0)
Clamp, 2015 [84]	Cross-sectional	10	0.0	30.0 (6.0)	33.2 (3.0)	10	0.0	25.0 (5.0)	22.3 (1.6)
Cremonini, 2006 [85]	Parallel RCT	25	80.0	35.0 (8.0)	36.0 (4.0)	13	100.0	34.0 (12.0)	22.0 (2.0)
Daghestani, 2009 [57]	Cross-sectional	45	100.0	26.5 (6.4)	35.9 (6.2)	77	100.0	23.5 (4.5)	22.3 (3.3)
Druce, 2005 [86]	Parallel RCT	12	67.0	33.4 (6.9)	31.9 (3.5)	12	67.0	24.7 (4.6)	20.5 (0.6)
El Helou, 2019 [58]	Crossover	15	0.0	21.7 (3.5)	35.1 (3.9)	15	0.0	20.1 (1.5)	22.0 (1.9)
English, 2002 [28]	Cross-sectional	10	30.0	42.8 (7.0)	42.8 (6.4)	13	60.0	32.0 (8.3)	22.5 (1.9)
Erdmann, 2005 [59]	Cross-sectional	128	36.0	45.0 (13.6)	33.8 (5.7)	56	80.0	30.0 (10.5)	22.0 (2.2)
Espelund, 2005 [87]	Cross-sectional	16	56.0	39.7 (12.4)	29.5 (4.0)	17	47.0	33.7 (14.0)	23.4 (2.5)
Frecka, 2008 [88]	Cross-sectional	5	40.0	32.0 (9.6)	32.2 (1.6)	7	57.0	24.3 (4.2)	23.3 (1.6)
Guo, 2007 [60]	Cross-sectional	14	50.0	59.4 (7.3)	30.1 (1.9)	16	50.0	54.4 (4.5)	21.6 (1.7)
Haltia, 2010 [89]	Crossover	13	38.0	27.0 (6.0)	33.0 (4.5)	12	50.0	26.0 (5.0)	21.7 (1.3)
Karcz-Socha, 2011 [45]	Cross-sectional	96	50.0	51.5 (6.5)	35.3 (2.9)	46	52.0	51.2 (6.5)	23.4 (1.5)
Kheirouri, 2017 [62]	Cross-sectional	37	100.0	37.2 (7.5)	23.1 (0.8)	40	100.0	35.2 (7.9)	31.4 (0.7)
Kiessl, 2017 [63]	Crossover	43	100.0	37.2 (7.5)	31.5 (1.8)	42	100.0	35.2 (7.9)	21.7 (2.0)
Kocak, 2011 [64]	Cross-sectional	22	100.0	NR	34.1 (4.0)	19	100.0	NR	25.2 (1.7)
Kolodziejski, 2018 [43]	Cross-sectional	15	100.0	53.0 (8.5)	39.8 (1.0)	15	100.0	58.6 (10.3)	22.3 (0.5)
Korek, 2013 [32]	Cross-sectional	19	90.0	42.2 (3.3)	34.7 (4.9)	17	89.0	42.9 (5.3)	21.1 (1.9)
Korner, 2005 [90]	Cross-sectional	12	100.0	NR	34.1 (1.8)	8	100.0	NR	21.6 (0.7)
Lambert, 2011 [91]	Crossover	11	NR	32.5 (6.5)	29.3 (0.6)	11	NR	28.8 (4.8)	21.3 (0.6)
Leonetti, 2003 [65]	Cross-sectional	8	70.0	20.8 (0.6)	35.9 (2.5)	10	50.0	21.5 (0.5)	23.0 (3.6)
Marzullo, 2006 [17]	Cross-sectional	10	50.0	32.4 (1.6)	43.0 (0.9)	6	50.0	31.7 (1.3)	21.8 (1.4)
Outeiriño-Blanco, 2011 [67]	Cross-sectional	23	100.0	39.8 (2.9)	38.8 (1.2)	13	100.0	34.4 (3.6)	22.3 (0.7)
Ozkan, 2009 [68]	Parallel RCT	21	71.0	37.4 (12.4)	24.8 (3.1)	10	60.0	36.1 (9.8)	37.0 (3.6)
Papandreou, 2017 [69]	Cross-sectional	7	NR	21.0 (2.1)	31.3 (2.1)	13	NR	21.0 (0.8)	20.6 (3.0)
Pavlatos, 2005 [70]	Parallel RCT	9	100.0	39.5 (14.5)	37.2 (8.4)	9	100.0	38.7 (14.1)	23.0 (2.1)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

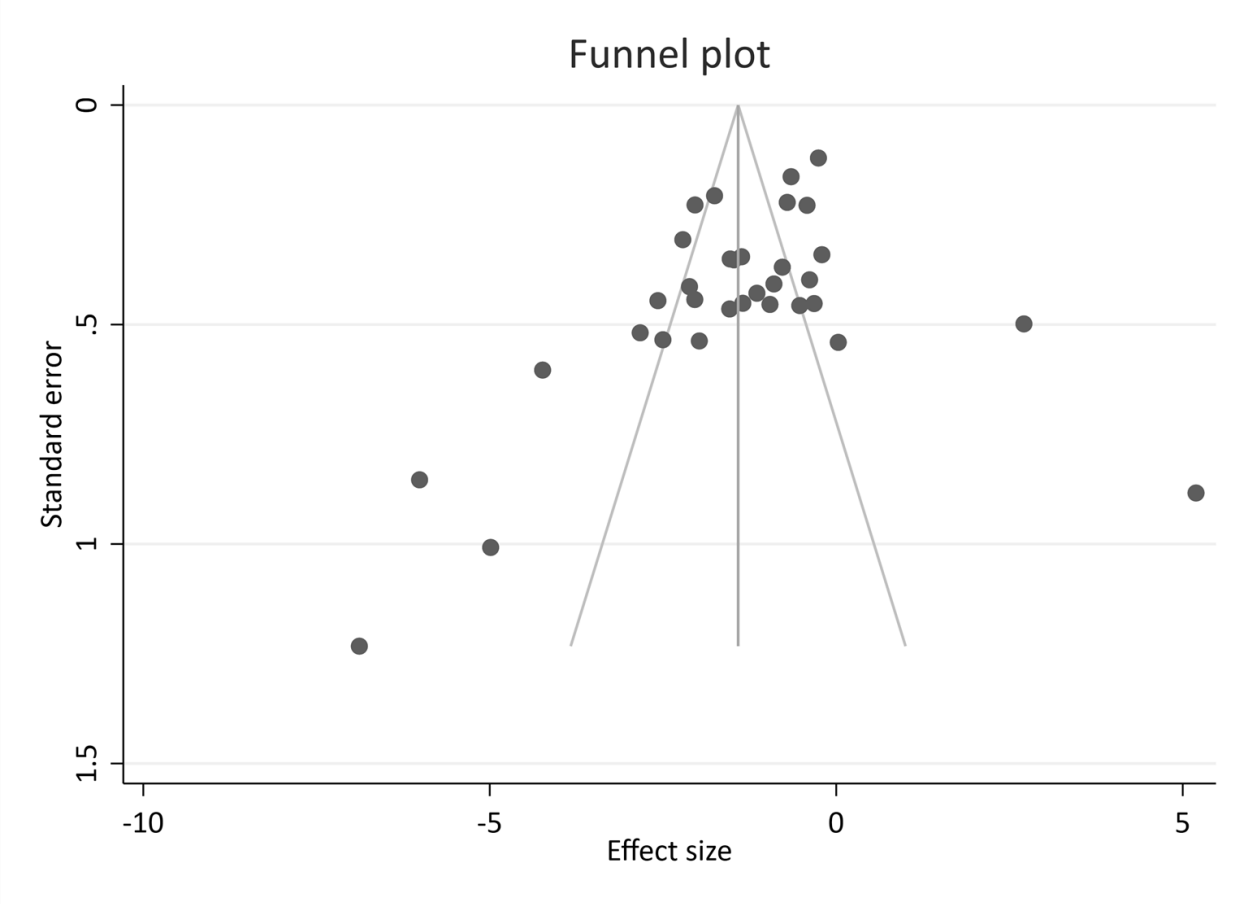

Figure 3: Funnel plot for total ghrelin (basal).

Active GLP-1 (basal)

Table 3: Summary of contributing articles for active GLP-1 (basal)

Author (year)	Study design	Obesity group n	Obesity group % female	Obesity group Age (years)	Obesity group BMI (kg/m²)	Control group n	Control group % female	Control group Age (years)	Control group BMI (kg/m²)
Adam, 2005 [94]	Crossover	28	68.0	44.4 (9.8)	30.4 (2.7)	30	50.0	31.6 (12.8)	22.9 (1.5)
Bowen, 2006 [52]	Crossover	47	0.0	56.8 (7.5)	30.1 (3.4)	25	0.0	50.5 (12.0)	23.3 (1.0)
Calanna, 2013 [95]	Cross-sectional	43	67.0	42.8 (13.1)	34.6 (3.9)	24	83.0	38.3 (9.8)	22.1 (2.0)
Carrol, 2007 [40]	Cross-sectional	20	55.0	47.2 (11.0)	36.8 (4.8)	19	47.0	39.2 (10.3)	22.6 (1.2)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Mersebach, 2003 [98]	Cross-sectional	16	75.0	39.3 (3.5)	36.1 (2.4)	10	70.0	37.9 (5.9)	21.5 (1.3)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

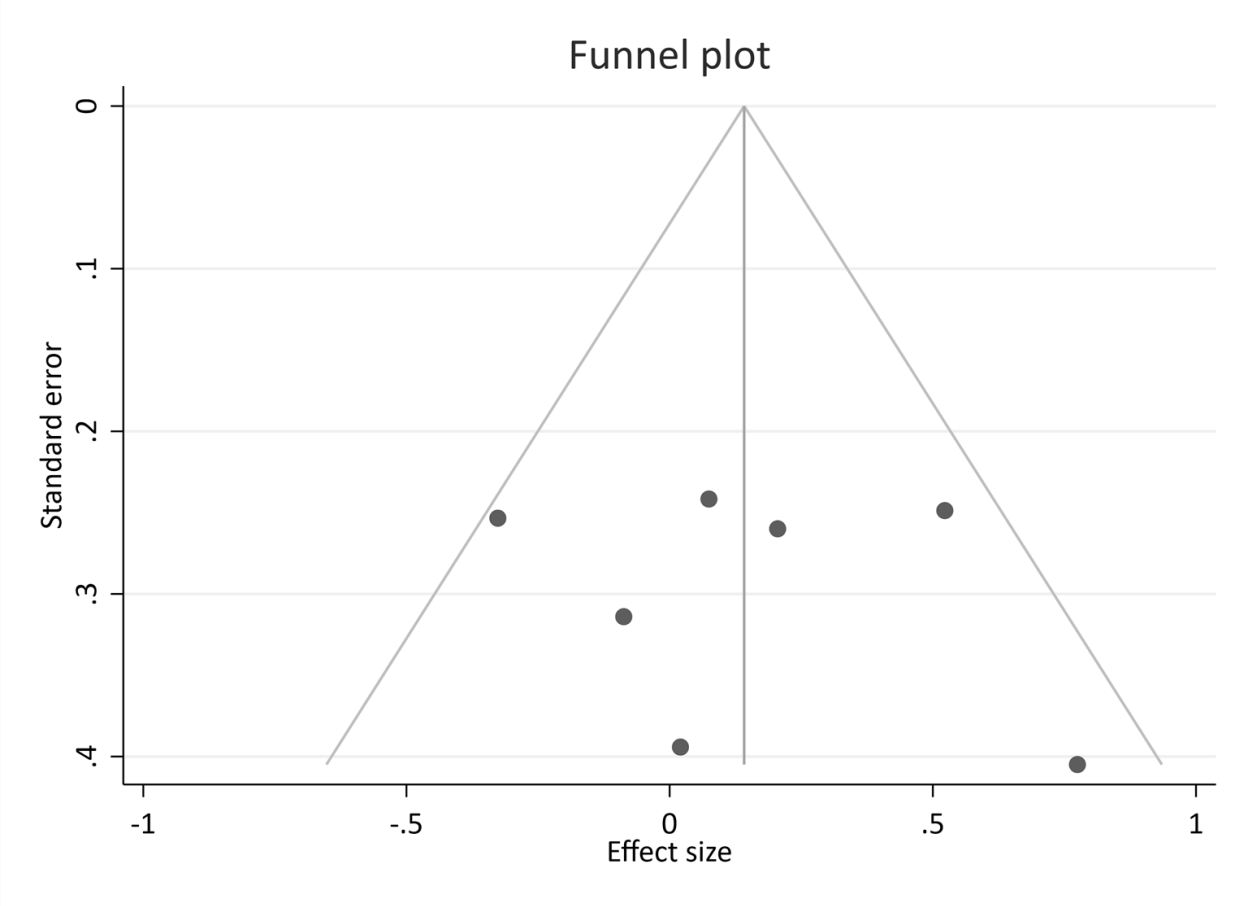

Figure 4: Funnel plot for active GLP-1 (basal).

Total GLP-1 (basal)

Table 4: Summary of contributing articles for total GLP-1 (basal)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Carr, 2010 [105]	Crossover	13	0.0	25.6 (4.0)	33.8 (0.6)	12	0.0	22.0 (1.8)	22.3 (0.3)
Cassar, 2015 [55]	Cross-sectional	18	100.0	35.0 (5.0)	31.0 (3.0)	22	100.0	28.0 (6.0)	22.0 (2.0)
Chia, 2017 [106]	Cross-sectional	20	50.0	68.5 (13.0)	35.6 (4.0)	20	50.0	68.5 (13.0)	23.6 (1.3)
Dirksen, 2019 [104]	Crossover	10	0.0	42.1 (9.8)	57.6 (17.2)	10	0.0	43.1 (8.9)	24.1 (3.8)
Douglas, 2017 [42]	Crossover	25	50.0	45.0 (12.4)	29.2 (2.9)	22	44.0	37.5 (15.2)	22.4 (1.5)
El Helou, 2019 [58]	Crossover	15	0.0	21.7 (3.5)	35.1 (3.9)	15	0.0	20.1 (1.5)	22.0 (1.9)
Elahi, 2016 [96]	Cross-sectional	12	50.0	42.0 (6.9)	37.2 (5.2)	12	50.0	29.0 (6.4)	22.3 (1.4)
Greenfield, 2009 [108]	Crossover	8	14.0	39.0 (9.8)	34.5 (4.4)	8	33.0	30.0 (5.8)	21.9 (2.2)
Haltia, 2010 [89]	Crossover	13	38.0	27.0 (6.0)	33.0 (4.5)	12	50.0	26.0 (5.0)	21.7 (1.3)
Heni, 2015 [109]	Cross-sectional	12	50.0	25.0 (6.9)	30.5 (6.2)	12	50.0	23.0 (6.9)	21.2 (3.8)
Nguyen, 2018 [110]	Cross-sectional	22	50.0	50.2 (2.5)	48.6 (1.8)	10	59.0	38.6 (8.4)	23.9 (0.7)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)
Verdich, 2001 [113]	.	19	0.0	35.0 (10.9)	38.1 (3.1)	12	0.0	24.2 (9.6)	23.1 (1.4)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

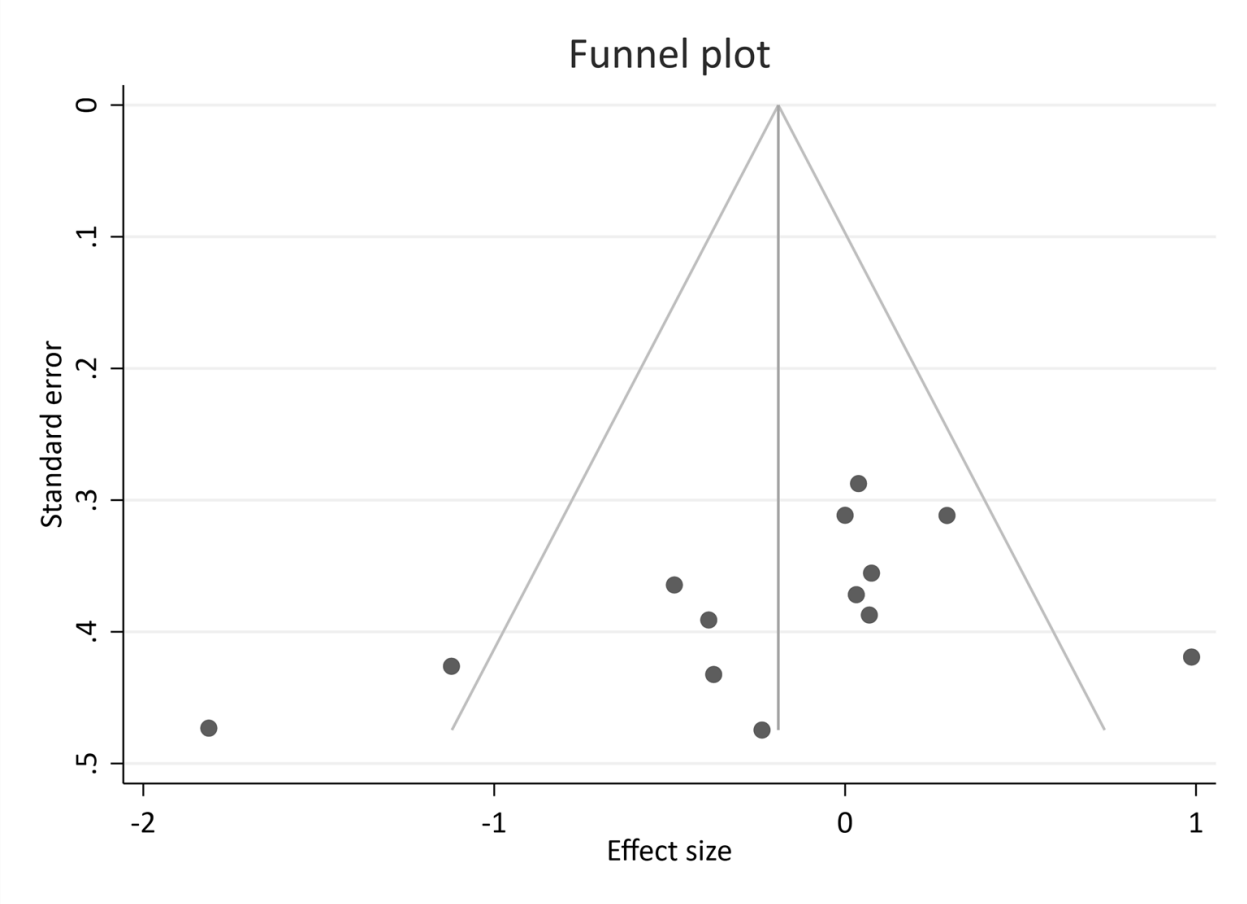

Figure 5: Funnel plot for total GLP-1 (basal).

Active PYY (basal)

Table 5: Summary of contributing articles for active PYY (basal)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Batterham, 2003 [49]	Crossover	12	50.0	29.0 (8.3)	33.0 (3.1)	12	50.0	27.3 (1.4)	20.5 (0.3)
Damgaard, 2013 [107]	Crossover	13	0.0	32.3 (7.7)	30.6 (3.5)	8	0.0	27.1 (6.1)	22.1 (2.3)
Dirksen, 2019 [104]	Crossover	10	0.0	42.1 (9.8)	57.6 (17.2)	10	0.0	43.1 (8.9)	24.1 (3.8)
Haltia, 2010 [89]	Crossover	13	38.0	27.0 (6.0)	33.0 (4.5)	12	50.0	26.0 (5.0)	21.7 (1.3)
Karcz-Socha, 2011 [45]	Cross-sectional	96	50.0	51.5 (6.5)	35.3 (2.9)	46	52.0	51.2 (6.5)	23.4 (1.5)
Korner, 2005 [90]	Cross-sectional	12	100.0	NR	34.1 (1.8)	8	100.0	NR	21.6 (0.7)
Outeiriño-Blanco, 2011 [67]	Cross-sectional	23	100.0	39.8 (2.9)	38.8 (1.2)	13	100.0	34.4 (3.6)	22.3 (0.7)
Pfluger, 2007 [117]	Crossover	15	100.0	52.0 (2.0)	31.1 (0.5)	17	100.0	51.6 (1.9)	22.0 (0.5)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

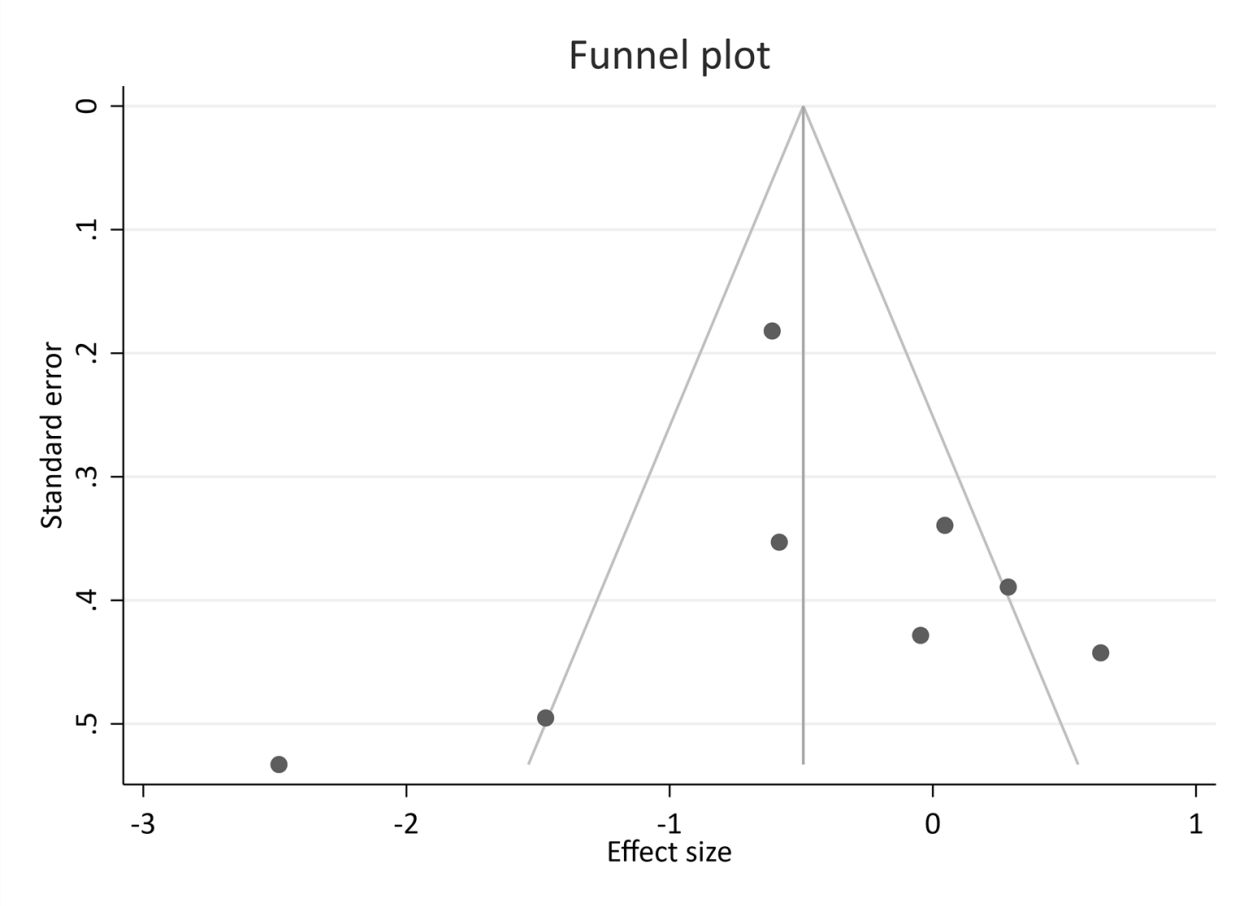

Figure 6: Funnel plot for active PYY (basal).

Total PYY (basal)

Table 6: Summary of contributing articles for total PYY (basal)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Brownley, 2010 [120]	Crossover	20	50.0	34.9 (9.0)	34.7 (2.8)	20	50.0	27.9 (7.0)	22.9 (1.4)
Cahill, 2011 [124]	Cross-sectional	28	0.0	23.2 (2.6)	29.1 (4.9)	41	0.0	23.1 (3.5)	23.1 (2.3)
Cahill, 2014 [123]	Cross-sectional	779	74.0	43.8 (12.1)	30.6 (4.8)	1315	76.0	42.1 (13.2)	24.4 (3.1)
Clamp, 2015 [84]	Cross-sectional	10	0.0	30.0 (6.0)	33.2 (3.0)	10	0.0	25.0 (5.0)	22.3 (1.6)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Douglas, 2017 [42]	Crossover	25	50.0	45.0 (12.4)	29.2 (2.9)	22	44.0	37.5 (15.2)	22.4 (1.5)
Pfluger, 2007 [117]	Crossover	79	73.0	47.6 (2.9)	35.1 (4.8)	66	70.0	41.5 (2.2)	22.1 (0.2)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

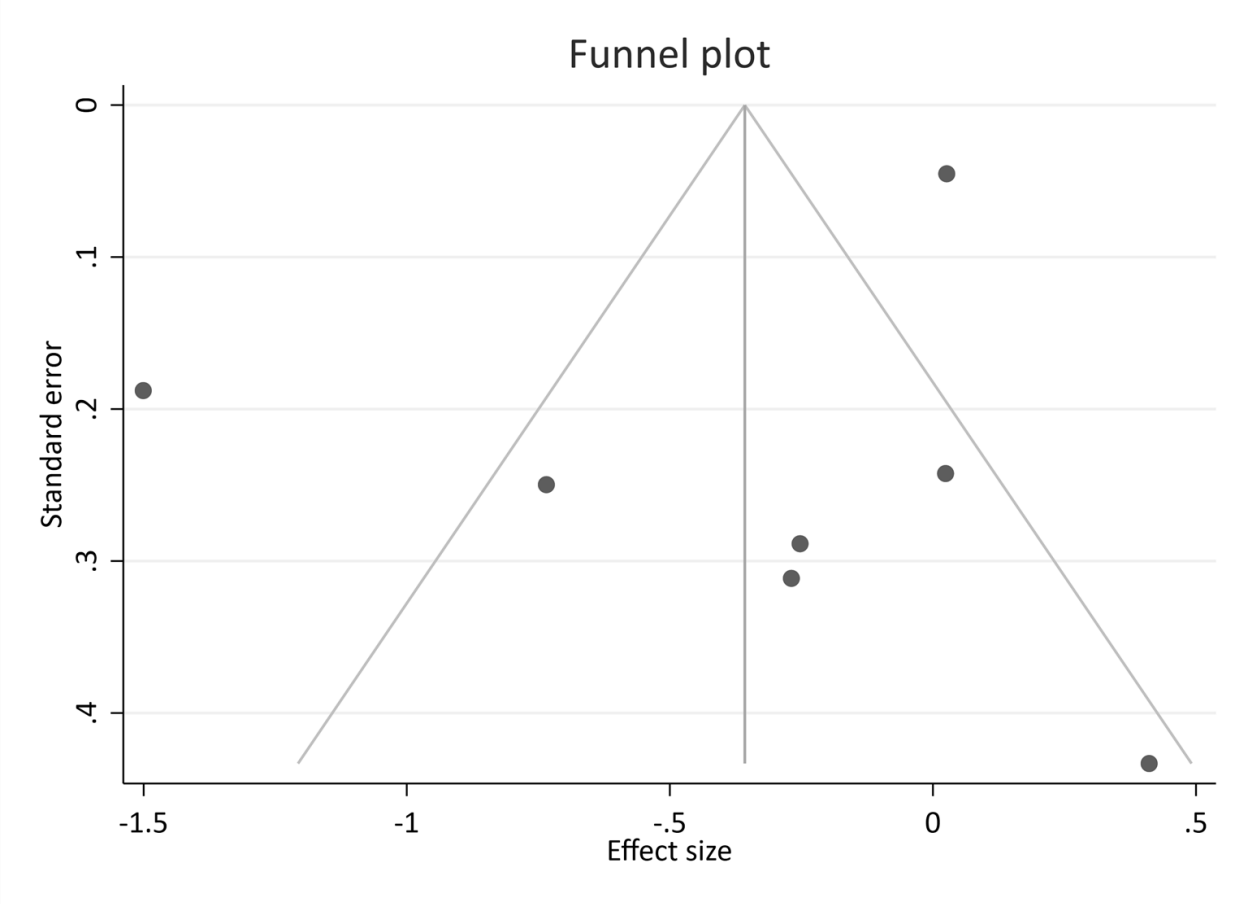

Figure 7: Funnel plot for total PYY (basal).

CKK (basal)

Table 7: Summary of contributing articles for CKK (basal)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Bowen, 2006 [52]	Crossover	47	0.0	56.8 (7.5)	30.1 (3.4)	25	0.0	50.5 (12.0)	23.3 (1.0)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Dirksen, 2019 [104]	Crossover	10	0.0	42.1 (9.8)	57.6 (17.2)	10	0.0	43.1 (8.9)	24.1 (3.8)
French, 1993 [18]	Crossover	8	63.0	NR	NR	7	57.0	NR	NR
Lieverse, 1993 [126]	Parallel RCT	7	100.0	40.2 (10.6)	40.7 (6.7)	7	100.0	41.2 (11.6)	22.3 (2.1)
Lieverse, 1998 [125]	Parallel RCT	7	100.0	NR	39.0 (2.0)	7	100.0	NR	22.0 (0.3)
Milewicz, 2000 [127]	Cross-sectional	25	NR	NR	34.6 (1.4)	16	NR	NR	21.2 (0.4)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

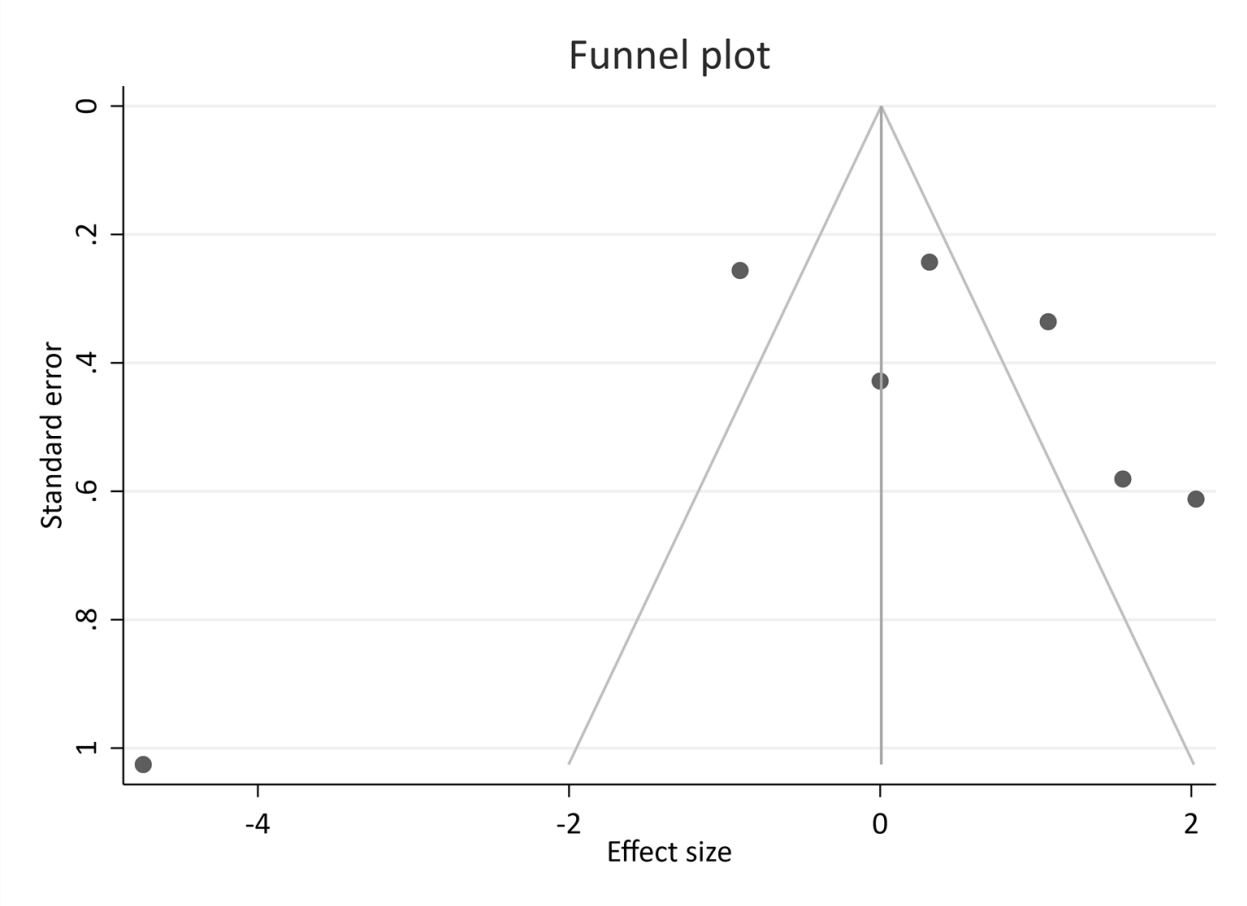

Figure 8: Funnel plot for CKK (basal)

Postprandial hormone concentrations

Total ghrelin (AUC)

Table 8: Summary of contributing articles for total ghrelin (AUC)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m²)	n	% female	Age (years)	BMI (kg/m²)
Brownley, 2012 [53]	Crossover	20	50.0	34.9 (9.0)	34.7 (2.8)	20	50.0	27.9 (7.0)	22.9 (1.4)
Frecka, 2008 [88]	Cross-sectional	5	40.0	32.0 (9.6)	32.2 (1.6)	7	57.0	24.3 (4.2)	23.3 (1.6)
Marzullo, 2006 [17]	Cross-sectional	10	50.0	32.4 (1.6)	43.0 (0.9)	6	50.0	31.7 (1.3)	21.8 (1.4)
Meyer-Gerspach, 2014 [15]	Cross-sectional	20	NR	29.8 (1.9)	39.3 (1.9)	20	NR	24.1 (0.6)	21.8 (0.4)
Outeiriño-Blanco, 2011 [67]	Cross-sectional	23	100.0	39.8 (2.9)	38.8 (1.2)	13	100.0	34.4 (3.6)	22.3 (0.7)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

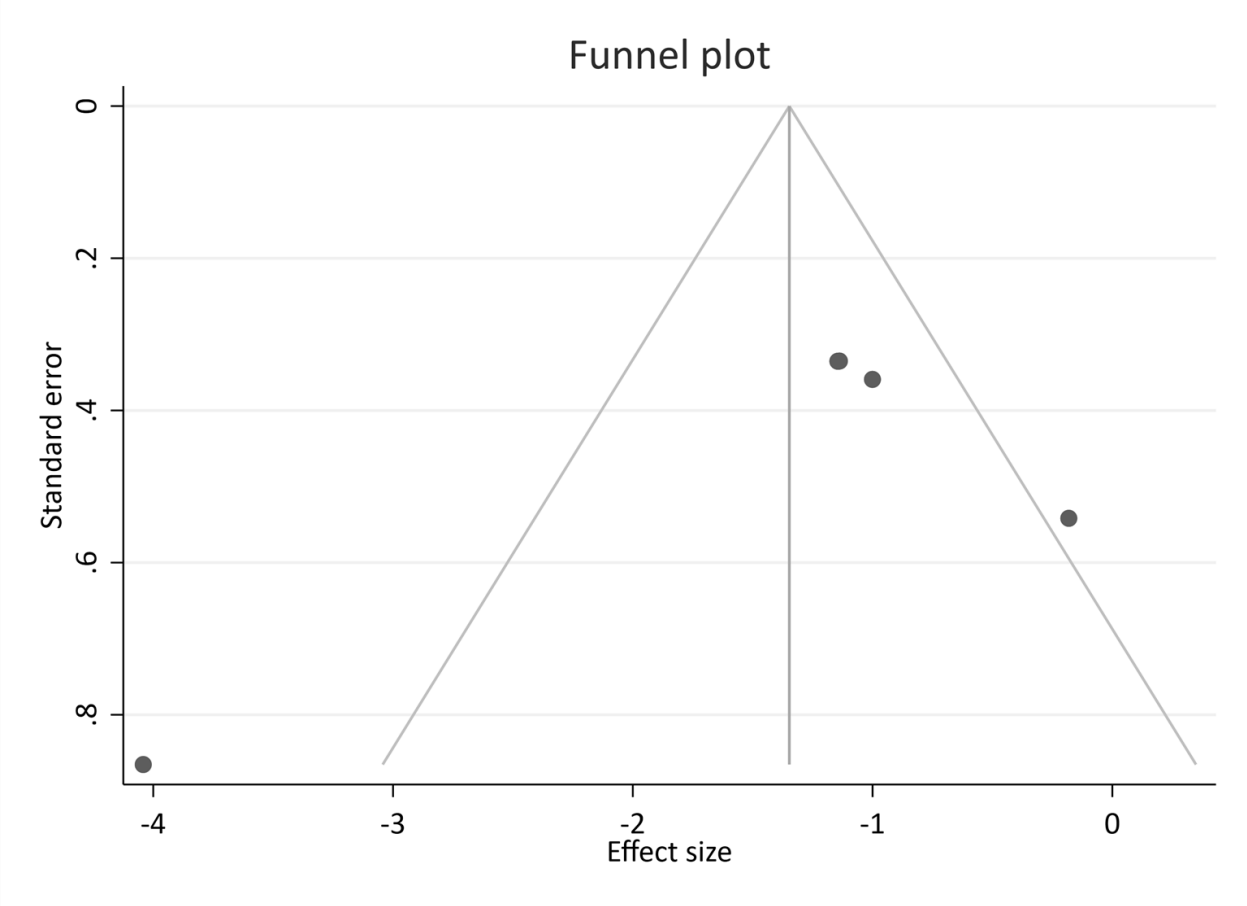

Figure 9: Funnel plot for total ghrelin (AUC).

Active GLP-1 (AUC)

Table 9: Summary of contributing articles for active GLP-1 (AUC)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Adam, 2005 [94]	Crossover	28	68.0	44.4 (9.8)	30.4 (2.7)	30	50.0	31.6 (12.8)	22.9 (1.5)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Heden, 2013 [97]	Crossover	13	54.0	25.4 (3.6)	34.6 (3.6)	13	46.0	26.0 (7.2)	23.0 (1.8)
Meyer-Gerspach, 2014 [15]	Cross-sectional	20	NR	29.8 (1.9)	39.3 (1.9)	20	NR	24.1 (0.6)	21.8 (0.4)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

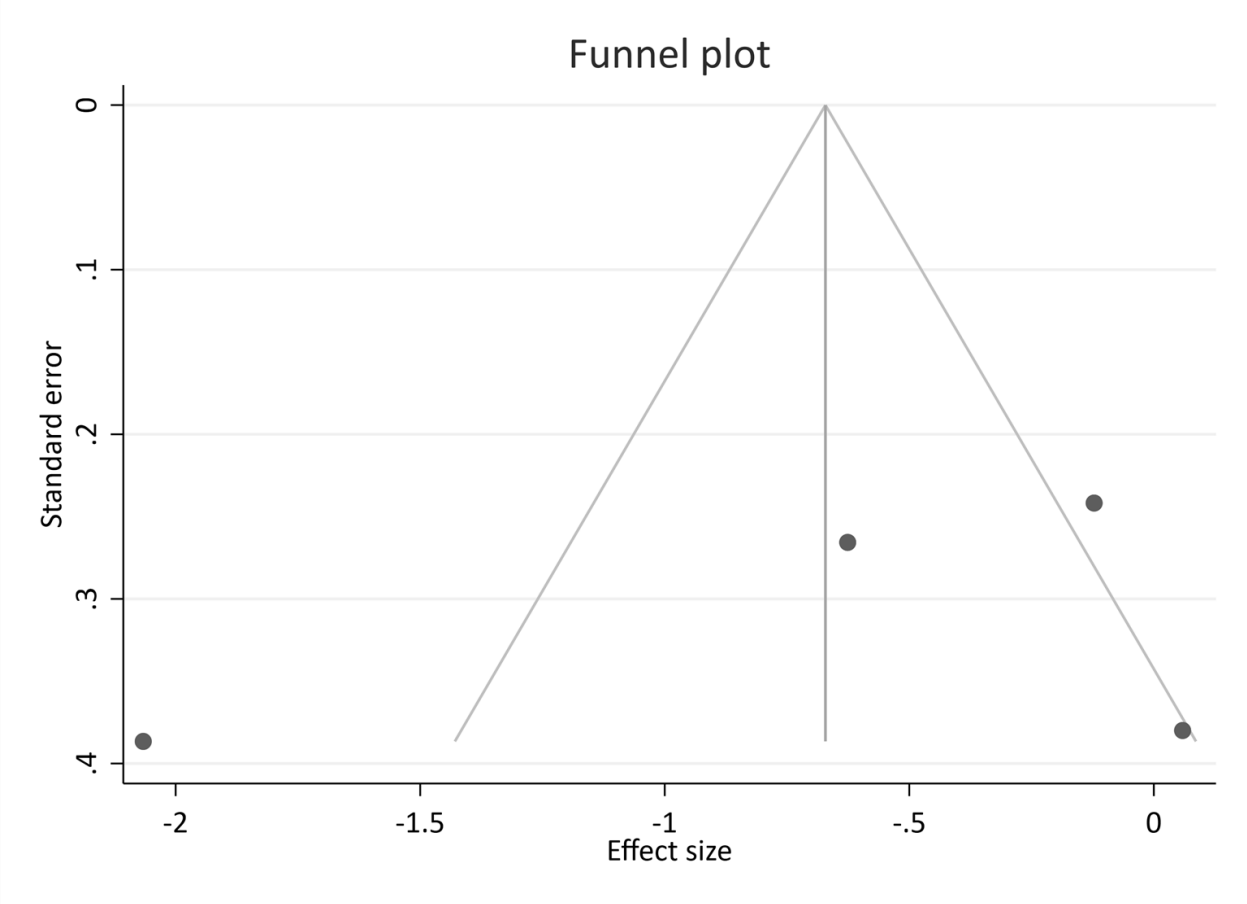

Figure 10: Funnel plot for active GLP-1 (AUC).

Total PYY (AUC)

Table 10: Summary of contributing articles for total PYY (AUC)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Brownley, 2010 [120]	Crossover	20	50.0	34.9 (9.0)	34.7 (2.8)	20	50.0	27.9 (7.0)	22.9 (1.4)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Meyer-Gerspach, 2014 [15]	Cross-sectional	20	NR	29.8 (1.9)	39.3 (1.9)	20	NR	24.1 (0.6)	21.8 (0.4)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

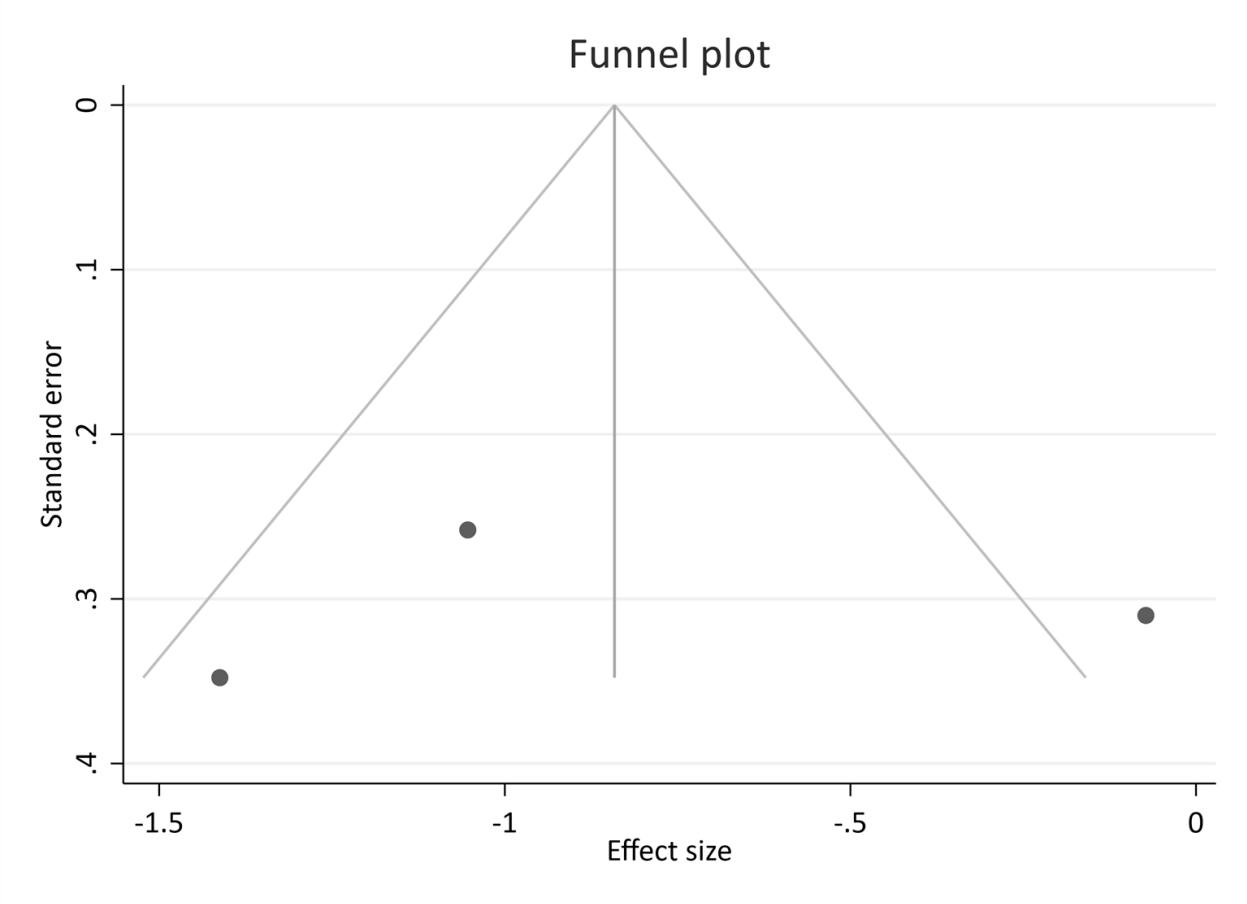

Figure 11: Funnel plot for total PYY (AUC).

Fasting appetite feelings

Hunger (fasting)

Table 11: Summary of contributing articles for hunger (fasting)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Andarini, 2017 [47]	Cross-sectional	16	0.0	21.4 (1.9)	33.6 (4.8)	16	0.0	20.6 (1.1)	21.3 (1.1)
Barkeling, 1995 [132]	Cross-sectional	38	50.0	43.5 (12.0)	39.4 (6.5)	38	50.0	40.0 (10.9)	22.3 (1.9)
Damgaard, 2013 [107]	Crossover	13	0.0	32.3 (7.7)	30.6 (3.5)	8	0.0	27.1 (6.1)	22.1 (2.3)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Dirksen, 2019 [104]	Crossover	10	0.0	42.1 (9.8)	57.6 (17.2)	10	0.0	43.1 (8.9)	24.1 (3.8)
El Helou, 2019 [58]	Crossover	15	0.0	21.7 (3.5)	35.1 (3.9)	15	0.0	20.1 (1.5)	22.0 (1.9)
Heden, 2013 [97]	Crossover	14	57.0	25.1 (5.0)	34.8 (4.4)	14	43.0	26.0 (6.0)	22.9 (1.7)
Iceta, 2019 [31]	Cross-sectional	55	100.0	38.0 (11.1)	41.5 (5.9)	29	100.0	37.0 (10.8)	21.5 (2.2)
Korner, 2005 [90]	Cross-sectional	12	100.0	NR	34.1 (1.8)	8	100.0	NR	21.6 (0.7)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

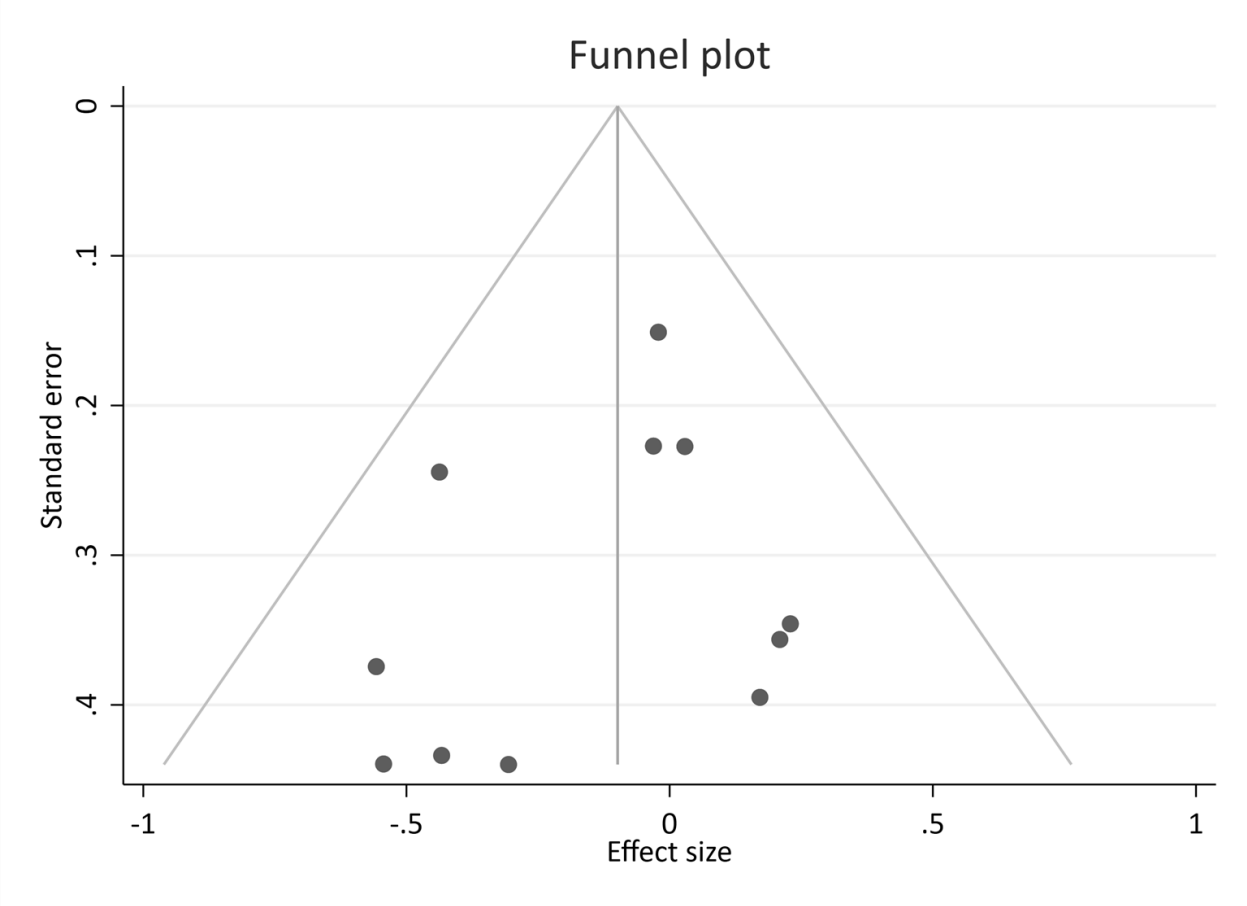

Figure 12: Funnel plot for hunger (fasting).

Fullness (fasting)

Table 12: Summary of contributing articles for fullness (fasting)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m²)	n	% female	Age (years)	BMI (kg/m²)
Barkeling, 1995 [132]	Cross-sectional	38	50.0	43.5 (12.0)	39.4 (6.5)	38	50.0	40.0 (10.9)	22.3 (1.9)
Damgaard, 2013 [107]	Crossover	13	0.0	32.3 (7.7)	30.6 (3.5)	8	0.0	27.1 (6.1)	22.1 (2.3)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Dirksen, 2019 [104]	Crossover	10	0.0	42.1 (9.8)	57.6 (17.2)	10	0.0	43.1 (8.9)	24.1 (3.8)
El Helou, 2019 [58]	Crossover	15	0.0	21.7 (3.5)	35.1 (3.9)	15	0.0	20.1 (1.5)	22.0 (1.9)
Heden, 2013 [97]	Crossover	14	57.0	25.1 (5.0)	34.8 (4.4)	14	43.0	26.0 (6.0)	22.9 (1.7)
Korner, 2005 [90]	Cross-sectional	12	100.0	NR	34.1 (1.8)	8	100.0	NR	21.6 (0.7)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

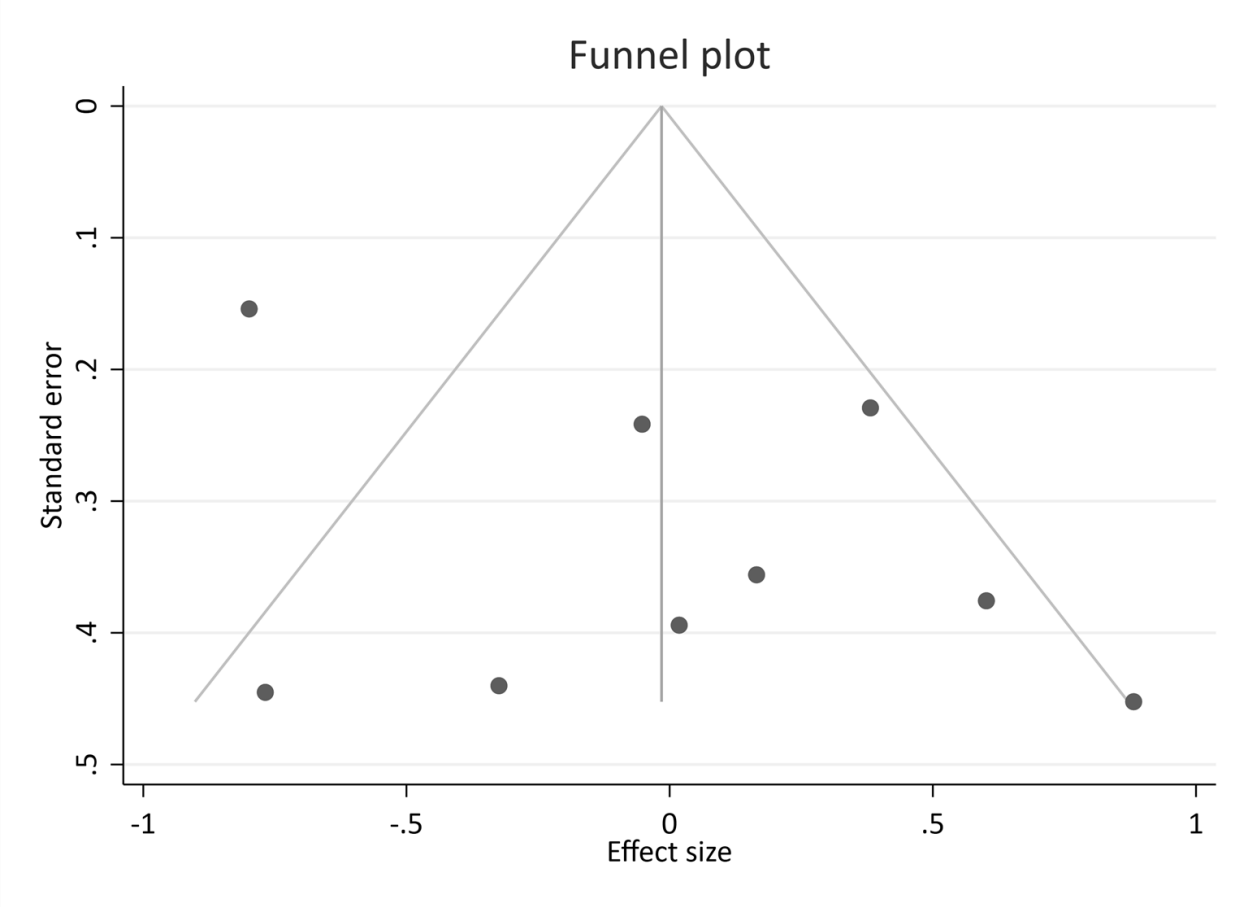

Figure 13: Funnel plot for fullness (fasting).

Desire to eat (DTE) (fasting)

Table 13: Summary of contributing articles for DTE (fasting)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m²)	n	% female	Age (years)	BMI (kg/m²)
Andarini, 2017 [47]	Cross-sectional	16	0.0	21.4 (1.9)	33.6 (4.8)	16	0.0	20.6 (1.1)	21.3 (1.1)
Barkeling, 1995 [132]	Cross-sectional	38	50.0	43.5 (12.0)	39.4 (6.5)	38	50.0	40.0 (10.9)	22.3 (1.9)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

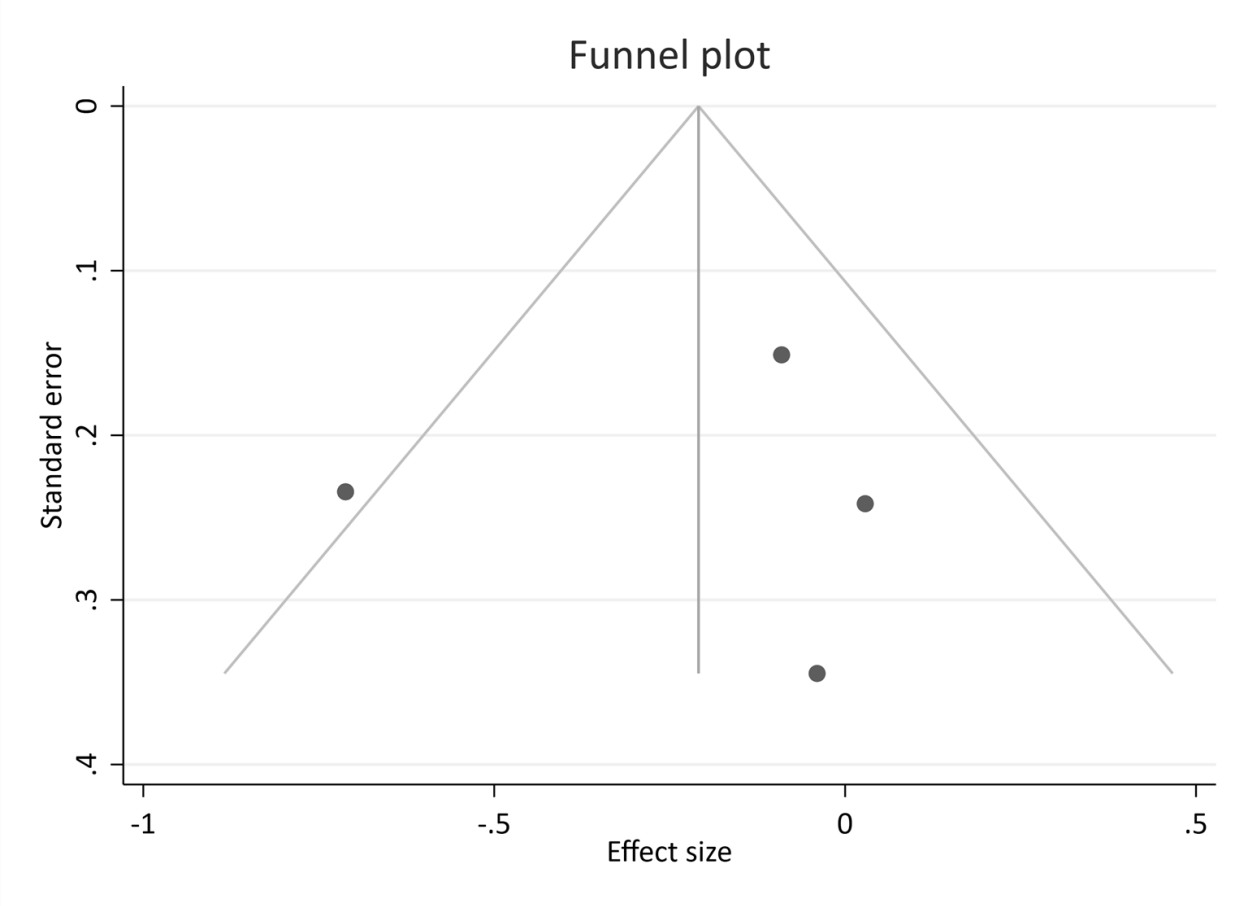

Figure 14: Funnel plot for DTE (fasting).

Prospective food consumption (PFC) (fasting)

Table 14: Summary of contributing articles for PFC (fasting)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m²)	n	% female	Age (years)	BMI (kg/m²)
Barkeling, 1995 [132]	Cross-sectional	38	50.0	43.5 (12.0)	39.4 (6.5)	38	50.0	40.0 (10.9)	22.3 (1.9)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
El Helou, 2019 [58]	Crossover	15	0.0	21.7 (3.5)	35.1 (3.9)	15	0.0	20.1 (1.5)	22.0 (1.9)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

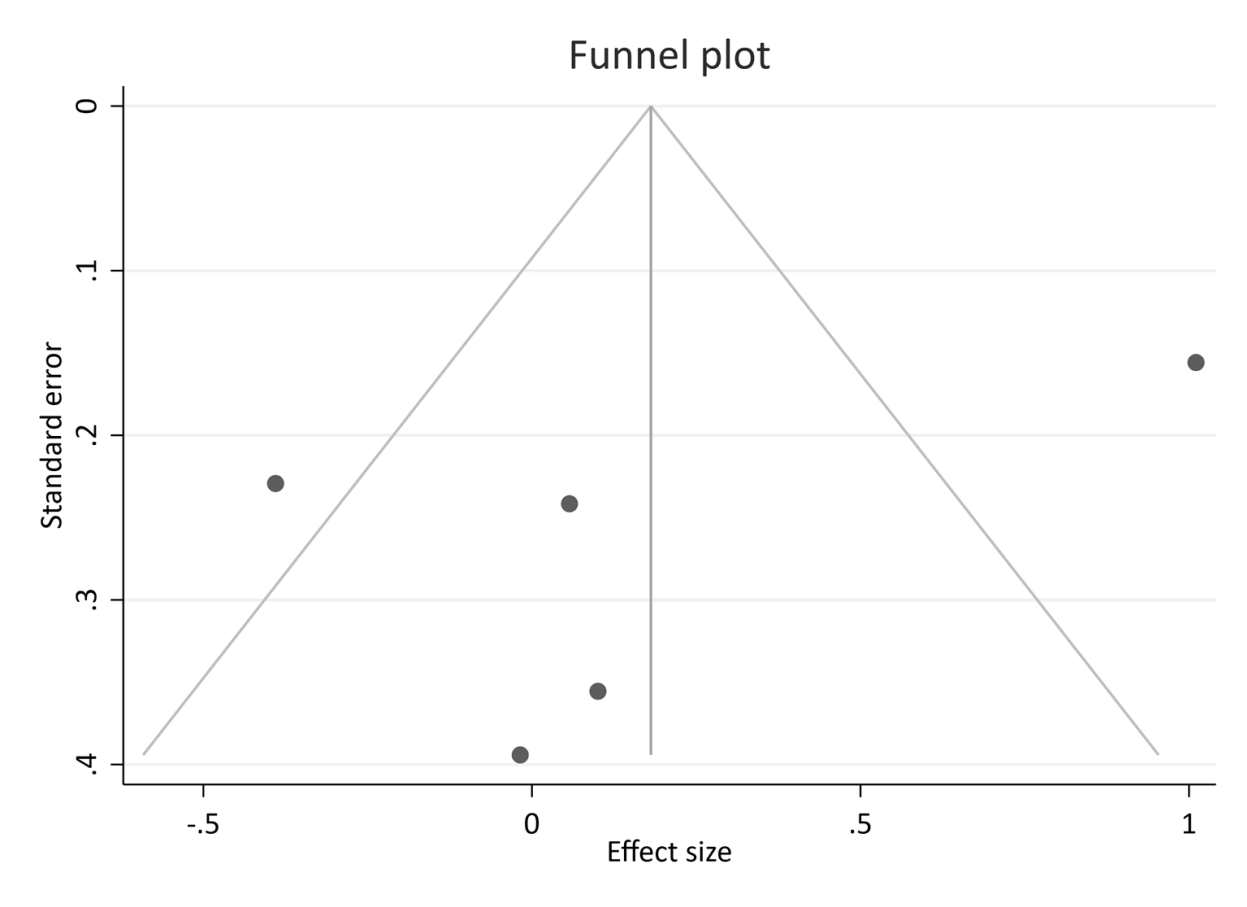

Figure 15: Funnel plot for PFC (fasting).

Postprandial appetite ratings

Hunger (AUC)

Table 15: Summary of contributing articles for hunger (AUC)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m²)	n	% female	Age (years)	BMI (kg/m²)
Andarini, 2017 [47]	Cross-sectional	16	0.0	21.4 (1.9)	33.6 (4.8)	16	0.0	20.6 (1.1)	21.3 (1.1)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Frecka, 2008 [88]	Cross-sectional	5	40.0	32.0 (9.6)	32.2 (1.6)	7	57.0	24.3 (4.2)	23.3 (1.6)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

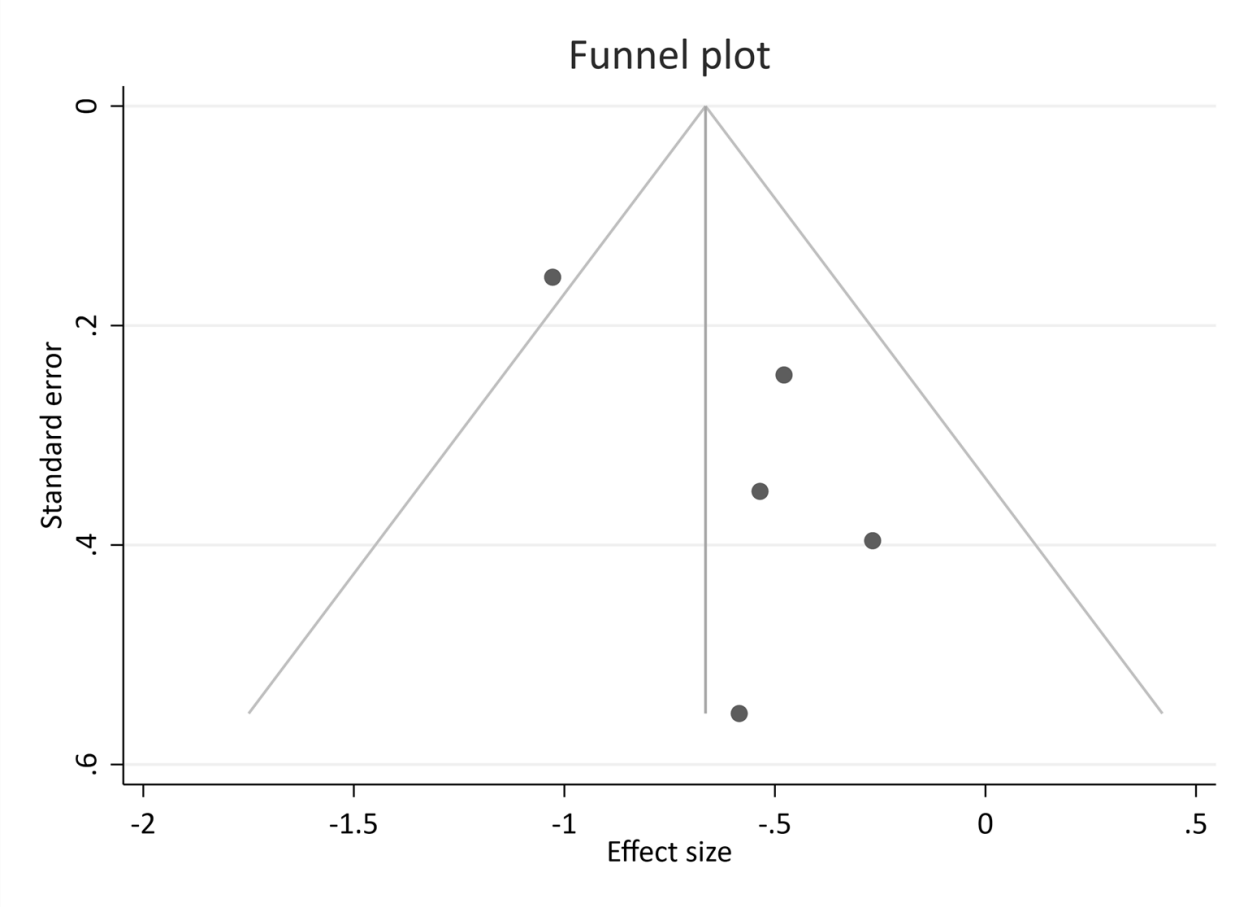

Figure 16: Funnel plot for hunger (AUC).

Fullness (AUC)

Table 16: Summary of contributing articles for fullness (AUC)

Author (year)	Study design	Obesity group n	Obesity group % female	Obesity group Age (years)	Obesity group BMI (kg/m²)	Control group n	Control group % female	Control group Age (years)	Control group BMI (kg/m²)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Frecka, 2008 [88]	Cross-sectional	5	40.0	32.0 (9.6)	32.2 (1.6)	7	57.0	24.3 (4.2)	23.3 (1.6)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

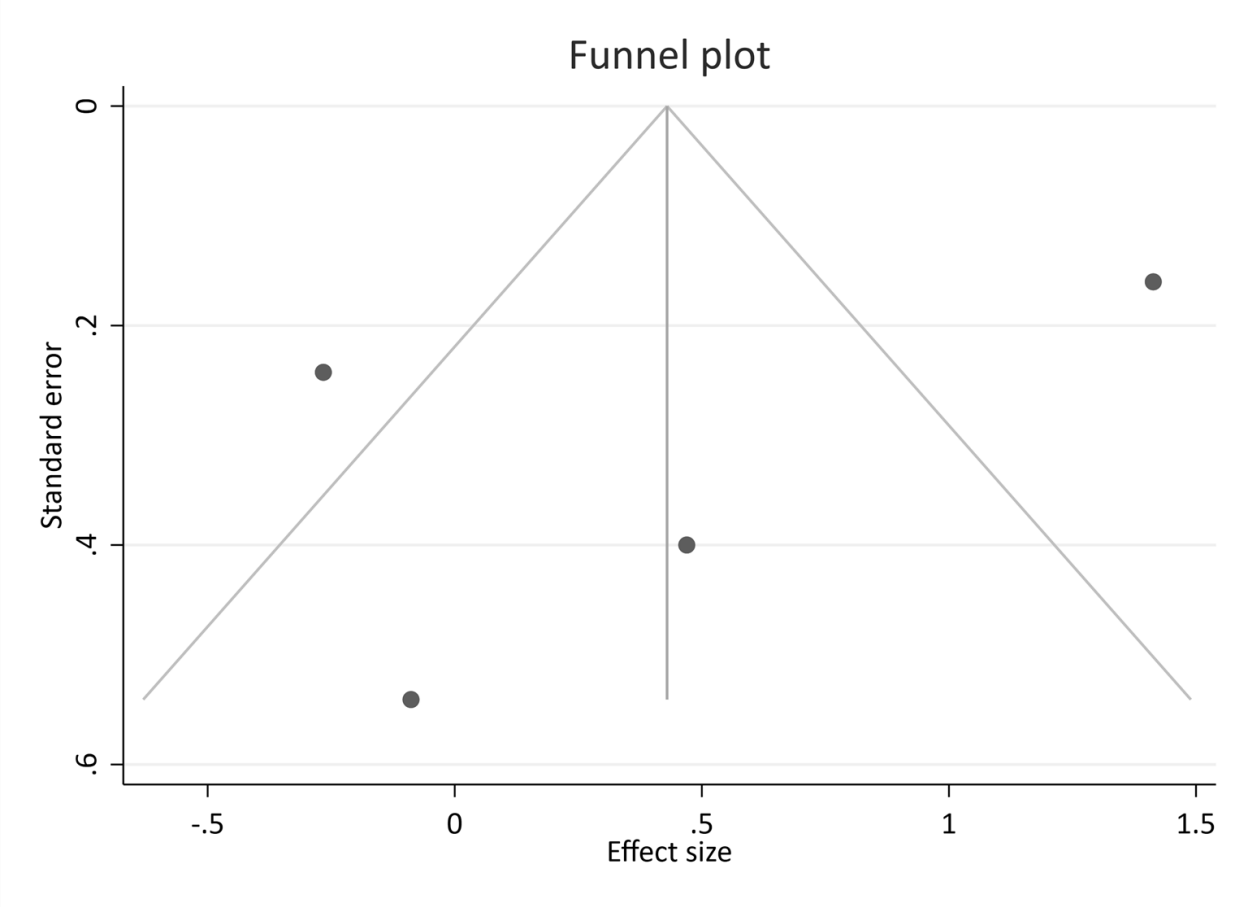

Figure 17: Funnel plot for fullness (AUC).

Desire to eat (DTE) (AUC)

Table 17: Summary of contributing articles for DTE (AUC)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
Andarini, 2017 [47]	Cross-sectional	16	0.0	21.4 (1.9)	33.6 (4.8)	16	0.0	20.6 (1.1)	21.3 (1.1)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Frecka, 2008 [88]	Cross-sectional	5	40.0	32.0 (9.6)	32.2 (1.6)	7	57.0	24.3 (4.2)	23.3 (1.6)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

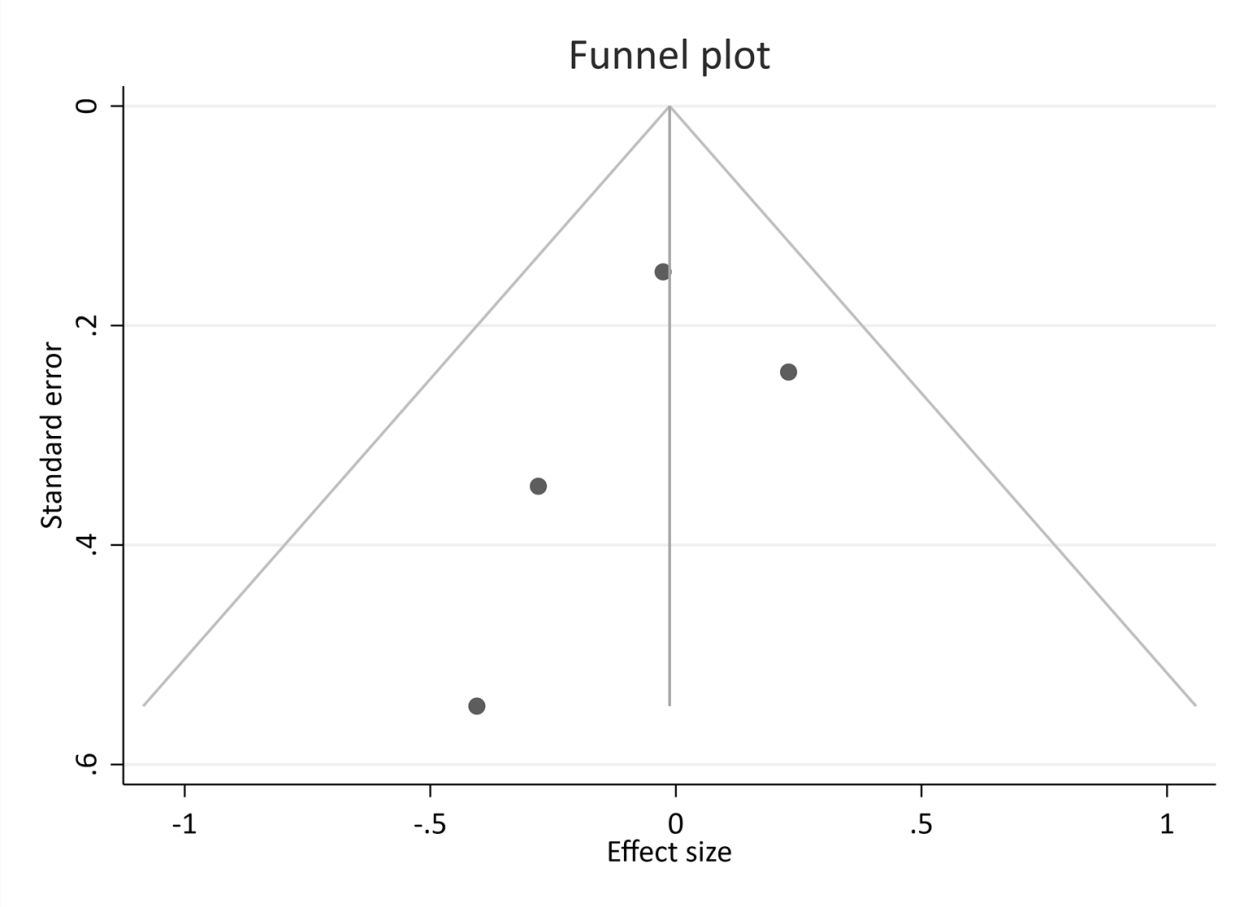

Figure 18: Funnel plot for DTE (AUC).

Prospective food consumption (PFC) (AUC)

Table 18: Summary of contributing articles for PFC (AUC)

Author (year)	Study design	Obesity group				Control group			
		n	% female	Age (years)	BMI (kg/m^2)	n	% female	Age (years)	BMI (kg/m^2)
DeBenedictis, 2020 [16]	Cross-sectional	34	50.0	38.9 (2.0)	34.0 (0.4)	33	49.0	45.0 (1.5)	24.8 (0.4)
Frecka, 2008 [88]	Cross-sectional	5	40.0	32.0 (9.6)	32.2 (1.6)	7	57.0	24.3 (4.2)	23.3 (1.6)
Painchaud Guerard, 2016 [135]	Parallel RCT	51	55.0	44.2 (16.2)	33.1 (3.6)	302	53.0	36.7 (14.7)	24.2 (2.9)
Smith, 2021 [103]	Crossover	12	0.0	34.8 (7.4)	33.7 (2.4)	12	0.0	35.8 (10.6)	23.7 (1.8)

BMI: Body mass index; NR: not reported; Values for age and BMI are presented as mean (SD)

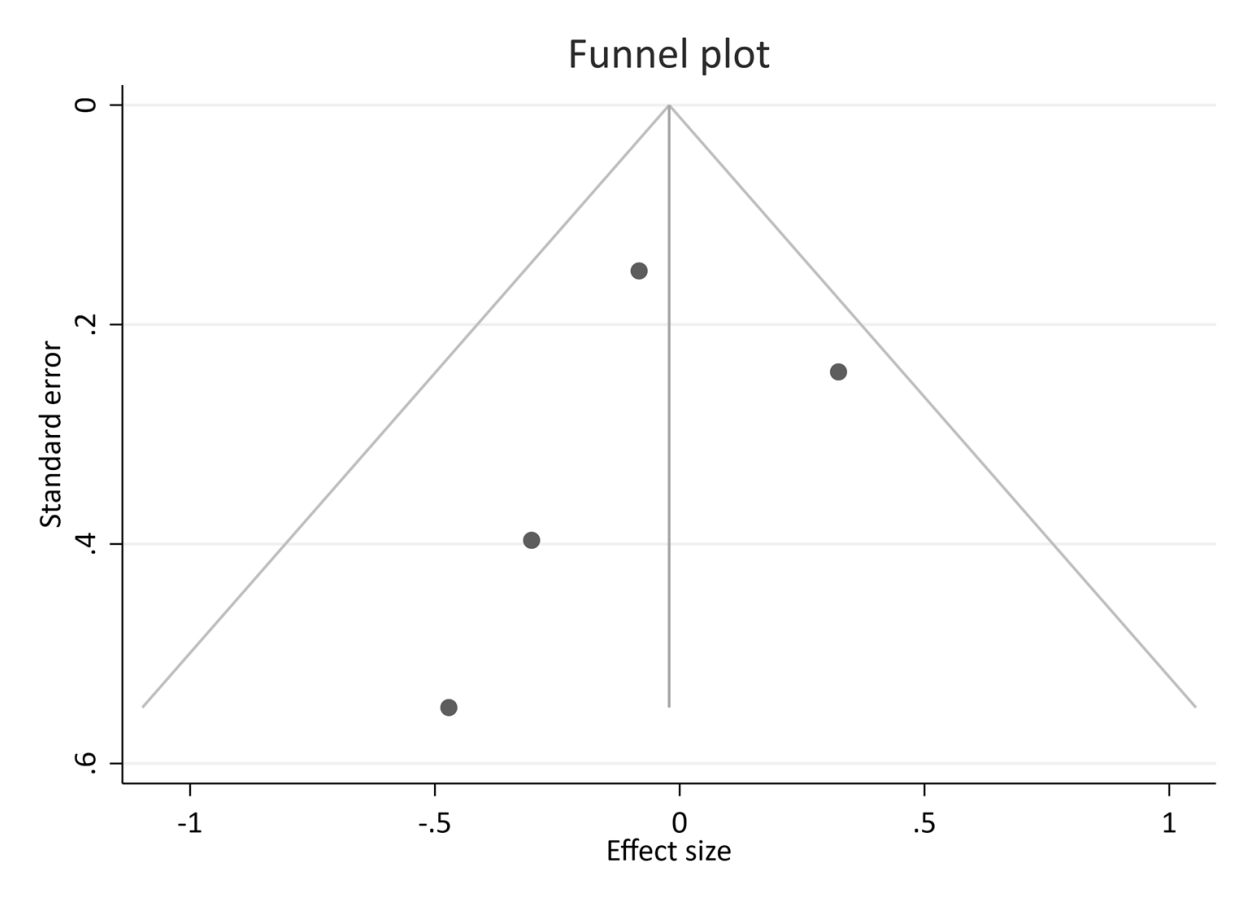

Figure 19: Funnel plot for PFC (AUC).

Supplementary table 19 – Risk of bias assessment

	Non-randomized trials							Randomized control trials					
Author (Year)	**1. Bias due to confounding**	**2. Bias in selection of participants into the study**	**3. Bias in classification of interventions**	**4. Bias due to deviations from intended interventions**	**5. Bias due to missing data**	**6. Bias in measurement of the outcome**	**7. Bias in selection of the reported result**	**1. Bias arising from the randomization process;**	**2. Bias due to deviations from intended interventions;**	**3. Bias due to missing outcome data;**	**4. Bias in measurement of the outcome**	**5. Bias in selection of the reported result**	**6. Overall**
Acosta (2015)	**low**	**moderate**	**low**	**moderate**	**moderate**	**low**	**low**						
Adam (2005)								**low**	**low**	**NA**	**moderate**	**low**	**moderate**
Andarini (2017)	**low**	**low**	**low**	**low**	**NA**	**moderate/high**	**low**						
Arafat (2013)								**low**	**low**	**NA**	**low**	**low**	**low**
Barkeling (1995)	**low**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Batterham (2003)								**low**	**low**	**NA**	**low**	**low**	**low**
Bogdanov (2019)	**low**	**low**	**low**	**moderate**	**moderate**	**moderate**	**low**						
Bowen (2006)								**low**	**low**	**NA**	**low**	**low**	**low**

Brownley (2010)	**moderate**	**moderate**	**low**	**low**	**moderate**	**low**	**low**						
Brownley (2012)	**low**	**moderate**	**low**	**moderate**	**NA**	**low**	**low**						
Cahill (2011)	**low**	**low**	**low**	**low**	**NA**	**moderate**	**low**						
Cahill (2014)	**low**	**low**	**low**	**low**	**moderate**	**moderate**	**low**						
Calanna (2013)	**moderate**	**low**	**low**	**low**	**NA**	**moderate**	**low**						
Carlson (2009)	**low**	**moderate**	**low**	**moderate**	**NA**	**low**	**low**						
Carr (2010)	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						
Carroll (2007)	**moderate**	**low**	**low**	**moderate**	**NA**	**low**	**low**						
Cassar (2015)								**low**	**low**	**NA**	**low**	**low**	**low**
Chearskul (2012)	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Chia (2017)								**low**	**low**	**NA**	**moderate**	**low**	**moderate**
Clamp (2015)	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Cremonini (2006)								**low**	**low**	**moderate**	**low**	**low**	**moderate**
Daghestani (2009)	**moderate**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Damgaard (2013)								**low**	**low**	**NA**	**low**	**low**	**low**
Dardzinska (2014)	**low**	**low**	**low**	**low**	**NA**	**moderate**	**low**						
DeBenedictis (2020)	**low**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Dirksen (2019)								**low**	**moderate**	**NA**	**low**	**low**	**moderate**

Douglas (2017)								**low**	**low**	**NA**	**modertae**	**low**	**moderate**
Druce (2005)								**moderate**	**low**	**NA**	**low**	**low**	**moderate**
El Helou (2019)								**low**	**low**	**NA**	**moderate**	**low**	**moderate**
Elahi (2016)	**moderate**	**low**	**low**	**low**	**NA**	**modertae**	**low**						
English (2002)	**moderate**	**low**	**low**	**low**	**NA**	**low**	**low**						
Erdmann (2005)	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Espelund (2005)	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						
Frecka (2008)	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
French (1993)								**low**	**low**	**NA**	**low**	**low**	**low**
Greenfield (2009)								**low**	**low**	**NA**	**low**	**low**	**low**
Guo (2007)	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Haltia (2010)								**moderate**	**moderate**	**NA**	**low**	**low**	**moderate**
Heden (2013)								**low**	**low**	**NA**	**moderate**	**low**	**moderate**
Heden (2013)								**low**	**low**	**NA**	**modertae**	**loe**	**moderate**
Heni (2015)	**low**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Homaee (2011)	**low**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Iceta (2019)	**low**	**moderate**	**low**	**low**	**low**	**moderate**	**low**						
Karcz-Socha et al. 2011	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						

Kheirouri et al. 2017								**low**	**moderate**	**NA**	**moderate**	**low**	**moderate**
Kiessl et al. 2017								**low**	**moderate**	**NA**	**moderate**	**low**	**moderate**
Kocak et al. 2011	**low**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Kolodziejski et al. 2018	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						
Korek et al. 2013	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Korner et al. 2005	**low**	**moderate**	**low**	**low**	**low**	**low**	**low**						
Krzyzanowska-Swinirska (2007)	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						
Lambert et al. 2011								**low**	**moderate**	**NA**	**high**	**low**	**moderate**
Leonetti et al.2003	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Lieverse et al. 1993								**low**	**low**	**NA**	**low**	**low**	**low**
Lieverse et al. 1998								**low**	**low**	**NA**	**low**	**low**	**low**
Lopez-Aguilar et al. 2018	**low**	**moderate**	**low**	**low**	**NA**	**moderate**	**low**						
Marzullo et al. 2004	**moderate**	**low**	**low**	**low**	**NA**	**low**	**low**						
Marzullo et al. 2006	**low**	**low**	**low**	**low**	**low**	**low**	**low**						
Mersebach et al. 2003	**low**	**low**	**low**	**low**	**NA**	**low**	**low**						
Meyer-Gerspach et al. 2014	**low**	**moderate**	**low**	**low**	**moderate**	**moderate**	**low**						
Milewicz et al. 2000	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						
Nguyen et al. 2018	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						

Outeiriño-Blanco et al. 2011	**low**	**moderate**	**low**	**low**	**NA**	**low**	**low**						
Ozkan et al. 2009								**low**	**low**	**NA**	**low**	**low**	**low**
Painchaud Guerard et al. 2016								**low**	**moderate**	**moderate**	**low**	**low**	**moderate**
Papandreou et al. 2017	**low**	**low**	**low**	**low**	**NA**	**moderate**	**low**						
Pavlatos et al. 2005								**low**	**low**	**NA**	**low**	**low**	**low**
Pfluger et al. 2007								**low**	**low**	**NA**	**moderate**	**low**	**moderate**
Smith et al. (2021)								**low**	**low**	**NA**	**moderate**	**low**	**moderate**
Verdich et al. (2001)	**low**	**low**	**low**	**moderate**	**NA**	**low**	**low**						

NA= not assessed/applicable

Supplementary table 20. Study characteristics

Study	Participants CO/OB	Test meal composition	Postprandial period	Hormone measure	Basal hormone concentrations CO vs OB	Postprandial hormone response CO vs OB (AUC)	Self-reported appetite ratings CO vs OB (VAS)
Acosta (2015) # (14)	BMI (kg/m^2): 27.6 ± 0.1 / 35.5 ± 3.9 Age: 37.3 ± 12.0 / 37.8 ± 12.1 N (F%): 105 (61%) / 201 (72%)			Total ghrelin [a *]	No difference P=0.063		
Adam (2005) (94)	BMI: 22.9 ± 1.5 / 30.4 ± 2.7 Age: 31.6 ± 12.8 / 44.4 ± 9.8 N (F%): 30 (50%) / 28 (68%)	454 kcal Protein (28.5%) CHO (48.8%) Fat (22.6%) Fiber (NA)	120 minutes 0, 30, 60, 90, 120	Active GLP-1[b *]	No difference P>0.05	CO > OB P=0.03	No differences in fullness or DTE for both fasting and postprandial (P>0.05)
Andarini (2017) (47)	BMI: 21.3 ± 1.1 / 33.6 ± 4.8 Age: 20.6 ± 1.1 / 21.4 ± 1.9 N (F%): 16 (0%) / 16 (0%)	570 kcal Protein (3.2%) CHO (14.8%) Fat (4.6%) Fiber (5.9 g)	120 minutes 0, 30, 60, 120	Active ghrelin [c*]	CO < OB P<0.01	CO < OB P<0.01	No differences in hunger or DTE for both fasting and postprandial (P>0.05)*
Arafat (2013) (26)	BMI: 21.7 ± 2.2 / 34.4 ± 5.6 Age: 25.1 ± 2.2 / 28.4 ± 8.6 N (F%): 13 (54%) / 11 (55%)			Active ghrelin [a*] Total ghrelin [a*]	CO > OB P<0.01, for both		
Barkeling (1995) # (132)	BMI: 22.3 ± 1.9 / 39.4 ± 6.5 Age: 40.0 ± 10.9 / 43.5 ± 12.0 N (F%): 38 (50%) / 38 (50%)						Fasting hunger, DTE, and PFC: CO>OB (P<0.0001) * Fasting fullness: CO<OB (P<0.0001) *
Batterham (2003) (49)	BMI: 20.5 ± 0.3 / 33.0 ± 3.1 Age: 27.3 ± 1.4 / 29.0 ± 8.3 N (F%): 12 (50%) / 12 (50%)			PYY 3-36 [d *] Total ghrelin [e*]	PYY CO > OB P<0.001 Ghrelin CO > OB P<0.001		
Bogdanov (2019) (51)	BMI: 21.8 ± 2.6 / 41.2 ± 2.1 Age: 37.0 ± 7.1 / 38.7 ± 9.4 N (F%): 15 (87%) / 15 (80%)			Total ghrelin [f*]	CO > OB P<0.001		
Bowen (2006) (52)	BMI: 23.3 ± 1.0 / 30.1 ± 3.4 Age: 50.5 ± 12.0 / 56.8 ± 7.5			Total ghrelin [g*] Active GLP-1 [h*]	Ghrelin CO > OB P<0.001		

	N (F%): 25 (0%) / 47 (0%)			CCK [i*]	GLP-1 CO < OB P<0.001 CCK no difference P>0.05	
Brownley (2010) # (120)	BMI: 22.9 ± 1.4 / 34.7 ± 2.8 Age: 27.9 ± 7.0 / 34.9 ± 8.9 N (F%): 20 (50%) / 20 (50%)	625 kcal Protein (14.7%) CHO (55.0%) Fat (30.2%) Fiber (NA)	180 minutes 0, 30, 60, 120, 180	Total PYY [a*]		
Brownley (2012) # (53)	BMI: 22.9 ± 1.4 / 34.7 ± 2.8 Age: 27.9 ± 7.0 / 34.9 ± 8.9 N (F%): 20 (50%) / 20 (50%)	625 kcal Protein (14.7%) CHO (55.0%) Fat (30.2%) Fiber (NA)	180 minutes -20, 30, 60, 90, 105, 120, 135, 150, 180	Total ghrelin [a*]	CO>OB P<0.001	CO> OB P<0.05
Cahill (2011) # (124)	BMI: 23.1 ± 2.3 / 29.1 ± 4.9 Age: 23.1 ± 3.5 / 23.2 ± 2.6 N (F%): 41 (0%) / 28 (0%)			Total PYY [f*]	No difference P>0.05	
Cahill (2014) # (122)	BMI: 24.4 ± 3.1 / 30.6 ± 4.8 Age: 42.2 ± 13.2 / 43.8 ± 12.1 N (F%): 1315 (76%) / 779 (74%)			Total PYY [f*]	No difference P>0.05	
Calanna (2013) (95)	BMI: 22.1 ± 2.0 / 34.6 ± 3.9 Age: 38.3 ± 9.8 / 42.8 ± 13.1 N (F%): 24 (83%) / 43 (67%)			Active GLP-1 [f*]	No difference P>0.05	
Carlson (2009) (54)	BMI: 23.1 ± 1.3 / 44.5 ± 7.1 Age: 32.2 ± 8.6 / 35.6 ± 9.7 N (F%): 10 (100%) / 13 (100%)			Total ghrelin [a*]	CO > OB P=0.001	
Carr (2010) (105)	BMI: 22.3 ± 0.3 / 33.8 ± 0.6 Age: 22.0 ± 1.8 / 25.6 ± 4.0 N (F%): 12 (0%) / 13 (0%)	560 kcal Macronutrient content of meal not specified.	300 minutes 0, 30, 45, 60, 90, 120, 150, 180, 240, 300	Total GLP-1 [j*]	No difference P>0.05	CO > OB P=0.022
Carroll (2007) # (40)	BMI: 22.6 ± 1.3 / 36.8 ± 4.8 Age: 39.2 ± 10.3 / 47.2 ± 10.9	379 kcal (528 ml) Protein (25.4 %)	60 minutes	Active ghrelin [a*] Active GLP-1 [a*]	No differences in ghrelin or GLP-1.	No differences in ghrelin or GLP-1.

	N (F%): 19 (47%) / 20 (55%)	CHO (42.3 %) Fat (7.6 %) Fiber (15.0g)	0, 10, 20, 30, 40, 50, 60		P>0.05	P>0.05	
Cassar (2015) (55)	BMI: 22.0 ± 2.0 / 31.0 ± 3.0 Age: 28.0 ± 6.0 / 35.0 ± 5.0 N (F%): 22 (100%) / 18 (100%)			Total ghrelin [k] Total GLP-1 [k]	Ghrelin CO > OB P<0.05. No difference in GLP-1 P>0.05		
Chearskul (2012) (41)	BMI: 20.7 ± 2.2 / 29.4 ± 3.6 Age: 30.1 ± 5.8 / 33.6 ± 6.3 N (F%): 53 (100%) / 33 (100%)			Active ghrelin [a*]	No difference P>0.05		
Chia (2017) (106)	BMI: 23.6 ± 1.3 / 35.6 ± 4.0 Age: 68.5 ± 13.0 / 68.5 ± 13.0 N (F%): 20 (50%) / 20 (50%)			Total GLP-1 [l*]	No difference P>0.05		
Clamp (2015) (84)	BMI: 22.3 ± 1.6 / 33.2 ± 3.0 Age: 25.0 ± 5.0 / 30.0 ± 6.0 N (F%): 10 (0%) / 10 (0%)			Total ghrelin [g*] Total PYY [a*]	No differences in ghrelin and PYY P>0.05		
Cremonini (2006) (85)	BMI: 22.0 ± 2.0 / 36.0 ± 4.0 Age: 34.0 ± 12.0 / 35.0 ± 8.0 N (F%): 13 (100%) / 25 (80%)			Total ghrelin [a*]	No difference P=0.068		
Daghestani (2009) # (57)	BMI: 22.3 ± 3.3 / 35.9 ± 6.2 Age: 23.5 ± 4.5 / 26.5 ± 6.4 N (F%): 77 (100%) / 45 (100%)	527 kcal Protein 21.5% CHO 54.4% Fat 24.1% Fiber (NA)		Total ghrelin [m*]	CO > OB P<0.001		
Damgaard (2013) # (107)	BMI: 22.1 ± 2.3 / 30.6 ± 3.5 Age: 27.1 ± 6.1 / 32.3 ± 7.7 N (F%): 8 (0%) / 13 (0%)		240 minutes Blood collection: 0, 7.5, 15, 22.5, 30, 37.5, 45, 52.5, 60, 90, 120, 150, 180, 210, 240 VAS: 0, 30, 60, 90, 120, 150, 180, 210, 240	Total GLP-1 [d] CCK [d] PYY 3-36 [n*]	No difference in GLP-1 or CCK P>0.05 No difference in PYY P= 0.15*		No differences in fasting hunger or postprandial hunger or fullness P>0.05 *

Dardzinska (2014) (27)	BMI: 23.0 ± 3.5 / 43.8 ± 6.8 Age: 37.2 ± 9.4 / 35.4 ± 9.1 N (F%): 12 (92%) / 12 (71%)	300 kcal (200 ml) Protein 16.0% CHO 49.0% Fat 35.0% Fiber (NA)	120 minutes Intervals not given	Active ghrelin [o*]	CO > OB P<0.00001	Ghrelin CO > OB P<0.05	
DeBenedictis (2020) (16)	BMI: 24.8 ± 0.4 / 34.0 ± 0.4 Age: 38.9 ± 2.0 / 45.0 ± 1.5 N (F%): 33 (49%) / 34 (50%)	600 kcal Protein (17%) CHO (48%) Fat (35%)	150 minutes Intervals not given	Active ghrelin [ø*] Total GLP-1 [r*] Total PYY [ø*] CCK [r*]		Ghrelin, GLP-1, PYY and CCK CO > OB P<0.001, for all	No differences in fasting or postprandial hunger, fullness, DTE or PFC * P > 0.05
Dirksen (2019) # (104)	BMI: 24.1 ± 3.8 / 57.6 ± 17.2 Age: 43.1 ± 8.9 / 42.1 ± 9.8 N (F%): 10 (0%) / 10 (0%)			Total GLP-1 [d*] PYY 3-36 [n*] CCK [d*]	GLP-1: CO > OB P<0.05 No differences in PYY or CCK P>0.05		No differences in fasting hunger or fullness P>0.05, for both*
Douglas (2017) (42)	BMI: 22.4 ± 1.5 / 29.2 ± 2.9 Age: 37.5 ± 15.2 / 45.0 ± 12.4 N (F%): 22 (44%) / 25 (50%)	610 kcal Protein 10.0 % CHO 72.0 % Fat 18.0 % Fiber (NA)		Active ghrelin [p*] Total GLP-1 [f*] Total PYY [f*]	No differences in ghrelin, GLP-1, or in PYY P>0.05, for all		No differences in hunger, fullness, or PFC (both in fasting and postprandially) P>0.05, for all
Druce (2005) (86)	BMI: 20.5 ± 0.6 / 31.9 ± 3.5 Age: 24.7 ± 4.6 / 33.4 ± 6.9 N (F%): 12 (50%) / 12 (50%)			Total Ghrelin [d*]	No difference. P>0.05		
El Helou (2019) (58)	BMI: 22.0 ± 1.9 / 35.1 ± 3.9 Age: 20.1 ± 1.5 / 21.7 ± 3.5 N (F%): 15 (0%) / 15 (0%)		240 minutes 0, 15, 30, 60, 120, 180, 240	Total ghrelin [q*] Total GLP-1 [f*]	Ghrelin: CO > OB P=0.050 GLP-1: No difference P>0.05		No differences in fasting hunger, fullness, or PFC P>0.05
Elahi (2016) (96)	BMI: 22.3 ± 1.4 / 37.2 ± 5.2 Age: 29.0 ± 6.4 / 42.0 ± 6.9 N (F%): 12 (50%) / 12 (50%)			Total GLP-1 [b*] Active GLP-1 [b]	total GLP-1 not compared No difference in active GLP-1		
English (2002) (28)	BMI: 22.5 ± 1.9 / 42.8 ± 6.4 Age: 32.0 ± 8.3 / 42.8 ± 7.0 N (F%): 13 (60%) / 10 (30%)	632 kcal Protein 12.2% CHO 56.8%	180 minutes 0, 15, 30, 60, 120, 180	Total ghrelin [d*]	CO > OB P=0.002		

		Fat 31.0% Fiber (NA)					
Erdmann (2005) (59)	BMI: 22.0 ± 2.2 / 33.8 ± 5.7 Age: 30.0 ± 10.5 / 45.0 ± 13.6 N (F%): 56 (80%) / 128 (36%)	260 kcal Protein 6.0% CHO 62.0% Fat 32.0% Fiber (NA)	180 minutes -15, 0, 15, 30, 60, 90, 120, 150, 180	Total ghrelin [g*]	CO > OB P<0.01		
Espelund (2005) (87)	BMI: 23.4 ± 2.5 / 29.5 ± 4.0 Age: 33.7 ± 14.0 / 39.7 ± 12.4 N (F%): 17 (47%) / 16 (56%)			Total ghrelin [r*]	No difference P>0.05		
Frecka (2008) (88)	BMI: 23.3 ± 1.6 / 32.2 ± 1.6 Age: 24.3 ± 4.2 / 32.0 ± 9.6 N (F%): 7 (57%) / 5 (40%)	501 ± 218 kcal Protein (11.1%) CHO (47.2%) Fat (29.7%)	330 minutes Intervals not given	Total ghrelin [a*]	No difference P>0.05	No difference P>0.05	No differences in fasting hunger, fullness, DTE, or PFC P>0.05, for all
French (1993) (18)	BMI: (20-25) / (<30) Age: (22-44) / (21-48) N (F%): 7 (57%) / 8 (63%)	317 kcal Macronutrient content of meal not specified	180 minutes 0, 10, 20, 30, 60, 90, 120, 150, 180	CCK [d*]	CO < OB P<0.01	CO < OB P<0.05	No differences in fasting hunger, fullness or DTE, P>0.05, for all. Postprandial hunger CO > OB P<0.05. No differences in postprandial fullness or DTE P>0.05, for both
Greenfield (2009) (108)	BMI: 21.9 ± 2.2 / 34.5 ± 4.4 Age: 30.0 ± 5.8 / 39.0 ± 9.8 N (F%): 7 (57%) / 8 (63%)			Total GLP-1 [d*]	No difference P>0.05		
Guo (2007) (60)	BMI: 21.6 ± 1.7 / 30.1 ± 1.9 Age: 54.4 ± 5.4 / 59.4 ± 7.3 N (F%): 16 (50%) / 14 (50%)	500 - 600 kcal Macronutrient content of meal not specified	Blood collection at 0 minutes	Total ghrelin [g*]	CO > OB P<0.01		
Haltia (2010) (89)	BMI: 21.7 ± 1.3 / 33.0 ± 4.5 Age: 26.0 ± 5.0 / 27.0 ± 6.0 N (F%): 12 (50%) / 13 (38%)			Total ghrelin [a*] PYY 3-36 [a*] Total GLP-1 [b*]	No differences in ghrelin, PYY or GLP-1 P>0.05, for all		

Heden (2013) (97)	BMI: 23.0 ± 1.8 / 34.6 ± 3.6 Age: 26.0 ± 7.2 / 25.4 ± 3.6 N (F%): 13 (46%) / 13 (54%)	600 kcal Protein (15%) CHO (45%) Fat (40%)	240 minutes 0, 5, 10, 15, 20, 30, 40, 50, 60, 75, 90, 120, 150, 180, 210, 240	Active GLP-1 [s*]		No difference P>0.05	
Heden (2013) (29)	BMI: 22.9 ± 1.7 / 34.8 ± 4.4 Age: 26.0 ± 6.0 / 25.1 ± 5.0 N (F%): 14 (43%) / 14 (57%)	600 kcal Protein (15%) CHO (45%) Fat (40%)	Blood collection at 0 minutes	Active ghrelin [s*]	No difference P=0.06		No differences in fasting hunger and fullness P>0.05, for both
Heni (2015) (109)	BMI: 21.2 ± 3.8 / 30.5 ± 6.2 Age: 23.0 ± 6.9 / 25.0 ± 6.9 N (F%): 12 (50%) / 12 (50%)			Total GLP-1 [t*]	No difference P>0.05		
Homaee (2011) (30)	BMI: 18.5 ± 2.2 / 31.0 ± 3.6 Age: 26.9 ± 5.6 / 27.5 ± 5.8 N (F%): 19 (0%) / 19 (0%)			Active ghrelin [u*]	CO > OB P=0.008		
Iceta (2019) (31)	BMI: 21.5 ± 2.2 / 41.5 ± 5.9 Age: 37.0 ± 10.8 / 38.0 ± 11.1 N (F%): 29 (100%) / 55 (100%)			Active ghrelin [o*]	CO > OB P<0.001		No differences in fasting hunger P>0.05
Karcz-Socha (2011) # (45)	BMI: 23.4 ± 1.5 / 35.7 ± 2.9 Age: 51.2 ± 6.5 / 51.5 ± 6.5 N (F%): 46 (52%) / 96 (50%)			Active ghrelin [a*] Total ghrelin [a*] PYY 3-36 [a*]	No difference in acylated ghrelin. Total ghrelin CO > OB P<0.001 PYY CO > OB P<0.001		
Kheirouri (2017) (62)	BMI: 23.1 ± 0.8 / 31.4 ± 0.7 Age: 35.2 ± 7.9 / 37.2 ± 7.5 N (F%): 40 (100%) / 37 (100%)			Total ghrelin [v*]	CO > OB P = 0.03		
Kiessl (2017) (63)	BMI: 21.7 ± 2.0 / 31.5 ± 1.8 Age: 18-30 / 18-30 N (F%): 42 (100%) / 43 (100%)	500 g pudding, eat as much as wanted. per100 g 158 kcal	60 minutes 0, 30, 60	Total ghrelin [f*]	CO > OB P<0.001	CO > OB P<0.001	
Kocak (2011) (64)	BMI: 25.2 ± 1.7 / 34.1 ± 4.0 Age: 58.6 ± 10.3 / 53.0 ± 8.5 N (F%): 19 (100%) / 22 (100%)			Total ghrelin [w*]	CO > OB P<0.001		

Kolodziejski (2018) (43)	BMI: 22.3 ± 0.5 / 39.8 ± 1.0 Age: 42.9 ± 5.3 / 42.2 ± 3.3 N (F%): 15 (100%) / 15 (100%)			Active ghrelin [n*] Total ghrelin [n*] Total GLP-1 [m]	Acylated ghrelin: CO < OB P<0.01 No difference for total ghrelin or GLP-1 P>0.05	
Korek (2013) (32)	BMI: 21.1 ± 1.85 / 34.7 ± 4.92 Age: 19-35 / 20-35 N (F%): 17 (89%) / 19 (90%)	260 kcal Protein (8%) CHO (43%) Fat (49%)	Blood collection at 0 and 120 minutes after breakfast	Active ghrelin [a*] Total ghrelin [n*]	CO > OB P<0.05	
Korner (2005) (90)	BMI: 21.6 ± 0.7 / 34.1 ± 1.8 Age: 30.6 ± 3.6 / 44.3 ± 4.3 N (F%): 8 (100%) / 12 (100%)	320 kcal Protein (35%) CHO (50%) Fat (15%)		Total ghrelin [g*] PYY 3-36 [g*]	No difference Ghrelin, P = 0.1 PYY, P>0.05	No differences in fasting hunger or fullness P>0.05
Krzyzanowska-Swinirska (2007) (46)	BMI: 21.3 ± 1.7 / 34.4 ± 4.1 Age: 28.8 ± 4.8 / 32.5 ± 6.5 N (F%): 32 (100%) / 30 (100%)			Active ghrelin [a*]	No difference	
Lambert (2011) # (91)	BMI: 21.3 ± 0.6 / 29.3 ± 0.6 Age: 21.5 ± 0.5 / 20.8 ± 0.6 N (F%): 11 (0%) / 11 (0%)			Total ghrelin [x*]	No difference	
Leonetti (2003) (65)	BMI: 23.0 ± 2.5 / 35.9 ± 3.6 Age: 40.2 ± 10.6 / 41.2 ± 11.6 N (F%): 10 (50%) / 8 (70%)			Total ghrelin [g*]	CO > OB P<0.01	
Lieverse (1993) (126)	BMI: 22.3 ± 2.1/ 40.7 ± 6.6 Age: 41.2 ± 11.6 / 40.2 ± 10.6 N (F%): 7 (100%) / 7 (100%)			CCK [d*]	No difference	
Lieverse (1998) (125)	BMI: 39 ± 2.0 / 22 ± 0.3 Age: 43 ± 3 / 42 ± 3 N (F%): 7 (100%) / 7 (100%)			CCK [d*]	CO > OB P<0.05	

Lopez-Aguilar (2018) (33)	BMI: 22.7 ± 1.5 / 35.4 ± 5.2 Age: 26.4 ± 5.6 / 29.2 ± 6.3 N (F%): 80 (67%) / 50 (52%)			Active ghrelin [v*]	CO > OB P=0.009	
Marzullo (2004) (35)	BMI: 22.4 ± 0.6 / 41.3 ± 1.1 Age: 31.7 ± 1.3 / 32.4 ± 1.6 N (F%): 20 (50%) / 20 (50%)			Active ghrelin [a*] Total ghrelin [g]	CO > OB P<0.05	
Marzullo (2006) (17)	BMI: 21.8 ± 1.4 / 43 ± 0.9 Age: 33.5 ± 2.4 / 31.8 ± 2.5 N (F%): 6 (50%) / 10 (50%)	500 kcal Liquid meal Proteins (17%) CHO (53%) Fat (30%)	120 minutes 0, 20, 40, 60, 80, 100, 120	Total ghrelin [g*]		CO > OB P<0.01
Mersebach (2003) (98)	BMI: 21.5 ± 1.3 / 36.1 ± 2.4 Age: 37.9 ± 5.9 / 39.3 ± 3.5 N (F%): 10 (70%) / 16 (75%)			Active GLP-1 [d*]	No difference	
Meyer-Gerspach (2014) (15)	BMI: 21.8 ± 0.4 / 39.3 ± 1.9 Age: 24.1 ± 0.6 / 29.8 ± 1.9 N (F%): 20 (NA%) / 20 (NA%)	750 kcal (500ml) Liquid meal Protein (17%) CHO (54%) Fat (29%)	180 minutes −1, 30, 60, 120, 180	Total ghrelin [a*] Active GLP-1 [f*] Total PYY [a*]	Ghrelin: CO > OB P = 0.027	Ghrelin: CO > OB P = 0.001 GLP-1: CO > OB P<0.001 PYY: CO > OB P<0.001
Milewicz (2000) # (127)	BMI: 21.2 ± 0.4 / 34.6 ± 1.4 Age: NA / NA N (F%): 16 (NA%) / 25 (NA%)			CCK [y*]	No difference	
Nguyen (2018) (110)	BMI: 23.9 ± 0.7 / 48.6 ± 1.8 Age: 38.6 ± 8.4 / 50.2 ± 2.5 N (F%): 10 (59%) / 22 (50%)	302 kcal 50 g minced beef 150 ml dextrose solution 75 g glucose loading	240 minutes −2, 15, 30, 45, 60, 75, 90, 120, 150, 180, 210, 240	Total GLP-1 [d*]	No difference	
Outeiriño-Blanco (2011) (67)	BMI: 22.3 ± 0.7 / 38.8 ± 1.2 Age: 34.4 ± 3.6 / 39.8 ± 2.9 N (F%): 13 (100%) / 23 (100%)	75 g glucose loading	150 minutes 0, 30, 60, 90, 120, 150	Total ghrelin [a*] PYY 1-36 [a*]	Ghrelin: CO > OB P= 0.01 PYY: no difference	Ghrelin: CO > OB P= 0.026 PYY: no difference

Ozkan (2009) (68)	BMI: 24.8 ± 3.1 / 37 ± 3.6 Age: 36.1 ± 9.8 / 37.4 ± 12.4 N (F%): 10 (60%) / 21 (71%)			Total ghrelin [z*]	CO > OB P= 0.001		
Painchaud Guerard (2016) # (135)	BMI: 24.2 ± 2.9 / 33.15 ± 3.6 Age: 36.7 ± 14.7 / 44.2 ±16.2 N (F%): 302 (53%) / 51 (55%)	Ad libitum oatmeal raisin snack	60 minutes -10, 0, 20, 40, 60				No differences in fasting or postprandial hunger, fullness, DTE or PFC *
Papandreou (2017) (69)	BMI: 20.6 ± 3 / 31.3 ± 2.1 Age: 21 ± 0.8 / 21 ± 2.07 N (F%): 13 (NA%) / 7 (NA%)			Total ghrelin [w*]	CO > OB P= 0.002		
Pavlatos (2005) (70)	BMI: 22.9 ± 2.1 / 37.2 ± 8.4 Age: 38.7 ± 14.1 / 39.5 ± 14.5 N (F%): 9 (100%) / 9 (100%)			Total ghrelin [a*]	CO > OB P= 0.001		
Pfluger (2007) # (117)	BMI: 22.1 ± 0.2 / 35.1 ± 4.8 Age: 41.5 ± 2.2 / 47.7 ± 2.9 N (F%): 66 (70%) / 79 (73%)			Total PYY [æ*]	No difference		
Pfluger (2007) # (117)	BMI: 22.0 ± 0.5 / 31.1 ± 0.5 Age: 51.6 ± 1.9 / 52.0 ± 2.0 N (F%): 17 (100%) / 15 (100%)			PYY 3-36 [a*]	No difference		
Smith (2021) # (103)	BMI: 23.7 ± 1.8 / 33.7 ± 2.4 Age: 35.8 ± 10.6 / 34.8 ± 7.4 N (F%): 12 (0%) / 12 (0%)	387 kcal Protein (15%) CHO (58%) Fat (27%)	240 minutes	Active GLP-1 [å*] Total GLP-1 [å*]	No difference in active GLP-1 Total GLP-1: CO<OB P <0.05		No differences in postprandial hunger, fullness, or PFC*
Verdich (2001) (113)	BMI: 23.1 ± 1.4 / 38.1 ± 3.1 Age: 24.2 ± 9.6 / 35.0 ± 10.9 N (F%): 12 (0%) / 19 (0%)	597 kcal Protein (20%) CHO (50%) Fat (30%)	180 minutes 0, 20, 40, 60, 80, 100, 120, 140, 160, 180	Total GLP-1[d*]		CO > OB P<0.01	

AUC: area under the curve. BMI: body mass index. CCK: cholecystokinin. CHO: carbohydrate. CO: control. DTE: desire to eat. GLP-1: glucagon-like peptide 1. NA: not assessed/compared. OB: obesity. PFC: prospective food consumption. PYY: peptide YY. VAS: visual analogue scale. Superscript letters denote hormone analyses method used: [a]RIA (Linco Research, Inc., St. Charles, MO). [b]ELISA (Linco, St. Charles, MO, USA). [c]ELISA (Elabscience, Biotechnology, Beijing). [d]RIA manufacturer not specified. [e] Hormone assay not specified. [f]ELISA (Millipore Corporation Pharmaceuticals, Billerica, MA). [g]RIA (Phoenix Pharmaceuticals, Belmont, CA). [h]Fluorescence immunoassay (Linco). [i]RIA (Euria-Diagnostica, Malmo, Sweden). [j]Amino terminal-specific assay (Linco Research). [k]Bio-Plex Pro Diabetes assay (CAT#171-A7001M; Biorad Laboratories, Hercules, CA, USA). [l]ELISA (Alpco Diagnostics, Salem, NH). [m]ELISA (Phoenix Pharma-ceuticals, INc (Belmont, CA)). [n]RIA (Millipore, Billerica, MA, USA). [o]Human Acylated Ghrelin EIA Kit (Biovendor, Czech Republic). [p]ELISA (SPI BIO, Montigny le Bretonneux, France). [q]ELISA (EMD Millipore Corp., St. Charles, Missouri). [r]«inhouse » RIA. [s]MILLIPLEX magnetic bead-based quantitative multiplex immunoassay with the MAGPIX instrumentation (Millipore, Billerica, MA). [t]ELISA (Millipore, Watford, UK). [u]ELISA (Acylated ghrelin Human ELISA, BioVendor, Germany. [v]ELISA, inspecified. [w]Immunochemilunometric assay, (IDS, SMBH, Germany). [x]ELISA (Phoenix Pharmaceuticals, Inc, Burlingame, CA). [y]RIA (Peninsula Lab., Belniont, CA). [z]RIA (Phoenix, Europe, Kalsruhe, Germany). [æ]ELISA (Diagnostic Systems Laboratories, Webster, TX). [ø]Human Metabolic Hormone Magnetic Bead Panel (LINCOplex Kit, Millipore, St Louis, MO). [å]ELISA (Merck Millipore). Stars * denote that data was included in meta-analysis. # denotes that group characteristics was merged for the purpose of this analysis.